SSM Optimization

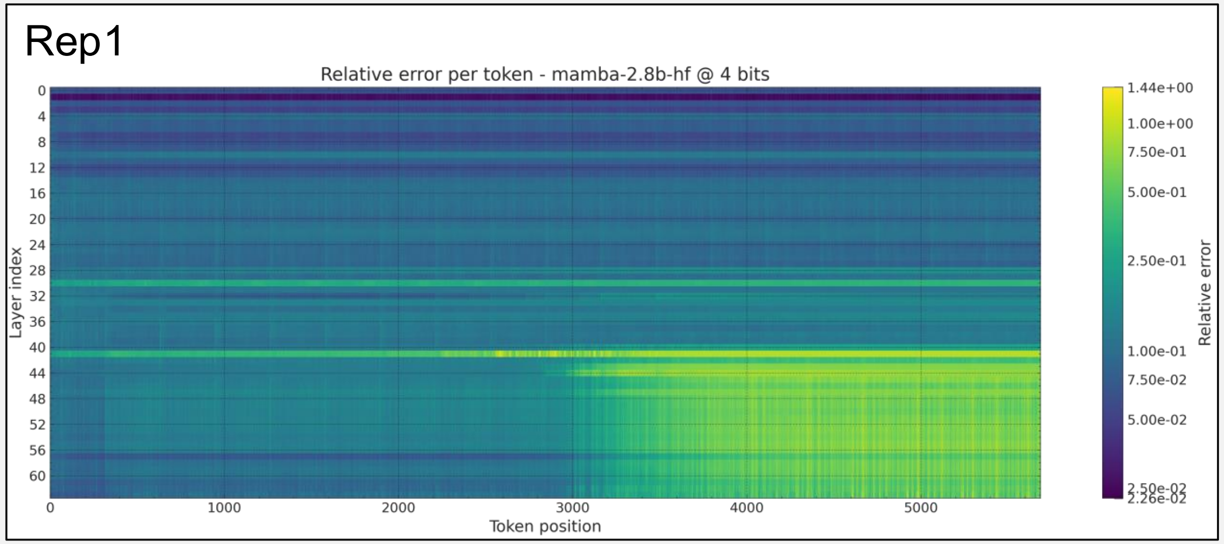
Weekly Meeting

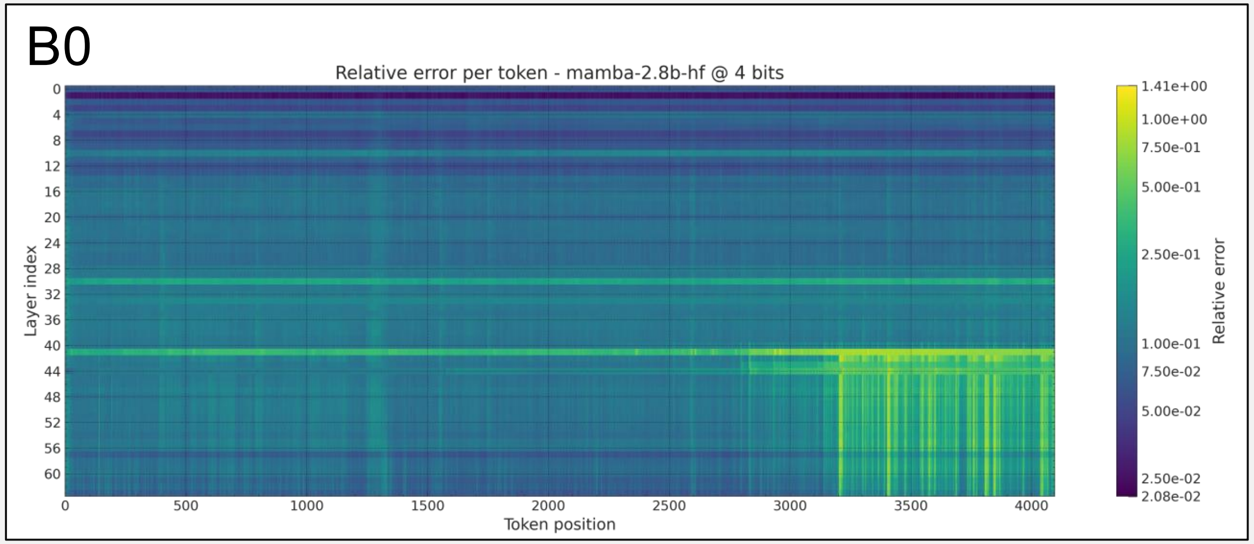
08/05/2026

Beatrice Alessandra Motetti

Skipping quantization of one layer

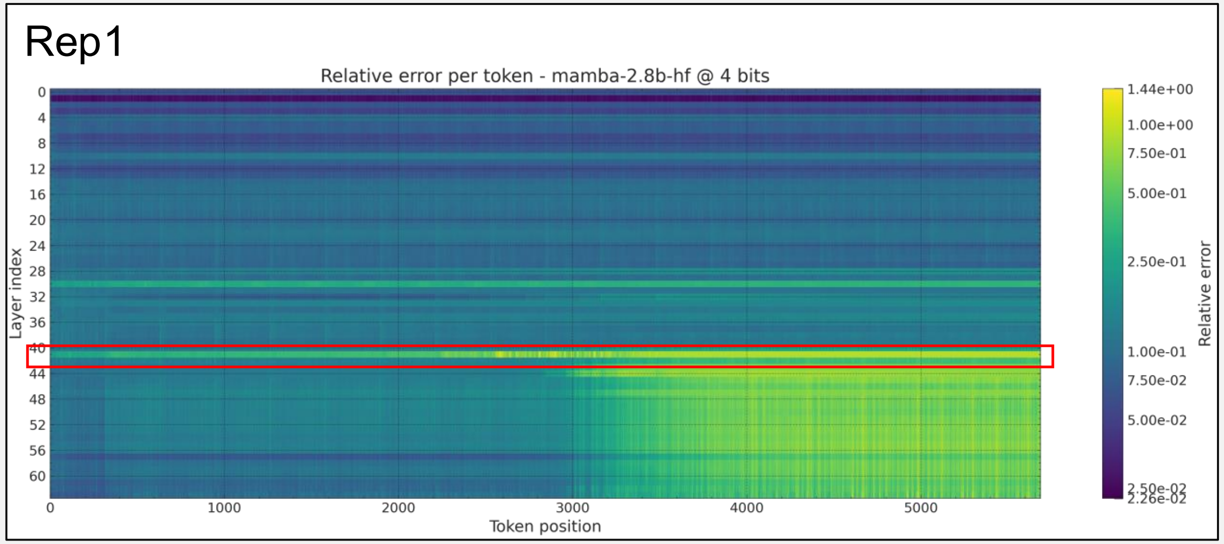
- From past observations:

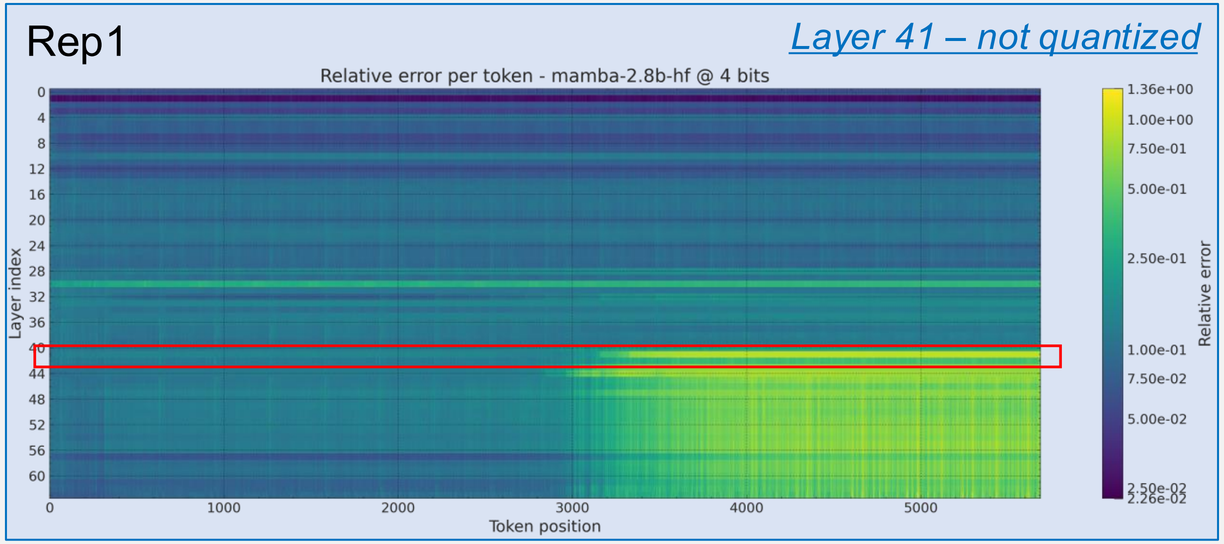




➢What is the effect of skipping quantization of layer 41?

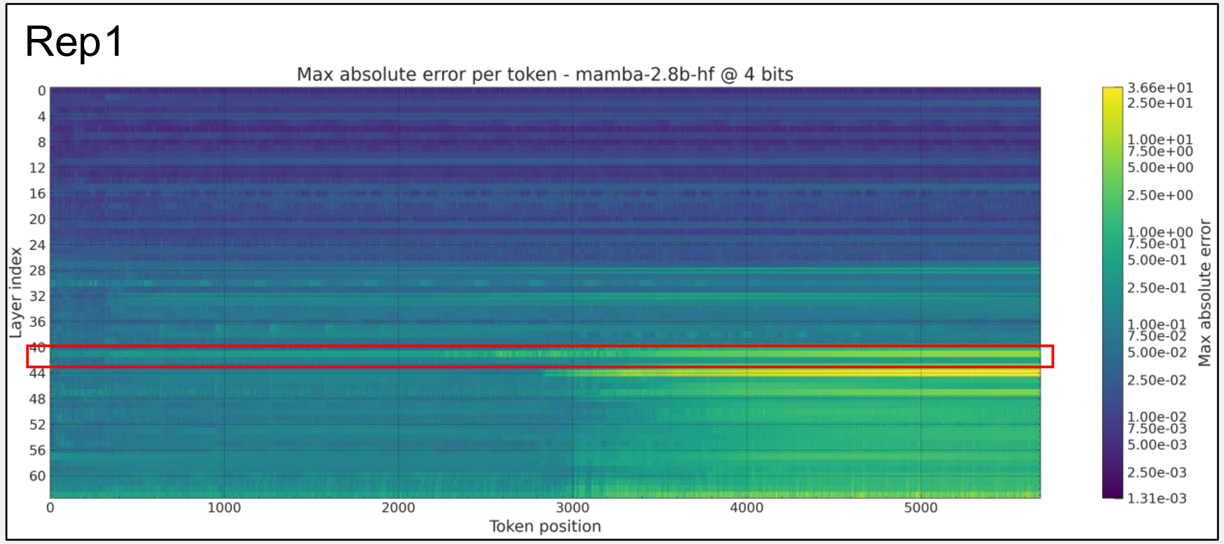
Skipping quantization of one layer

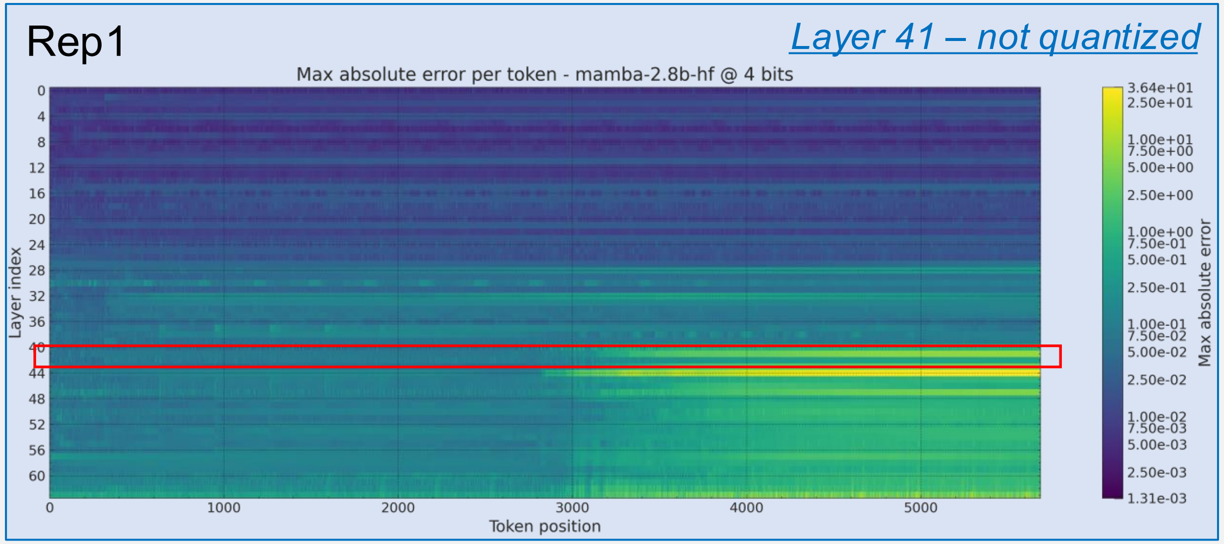




- The difference of the state with and without quantization for layer 41 decreases over the first timesteps (~0.4 -> ~0.2).
- From position ~3000, the difference still remains
  - (~1 → the difference is as large as the floating point state itself)

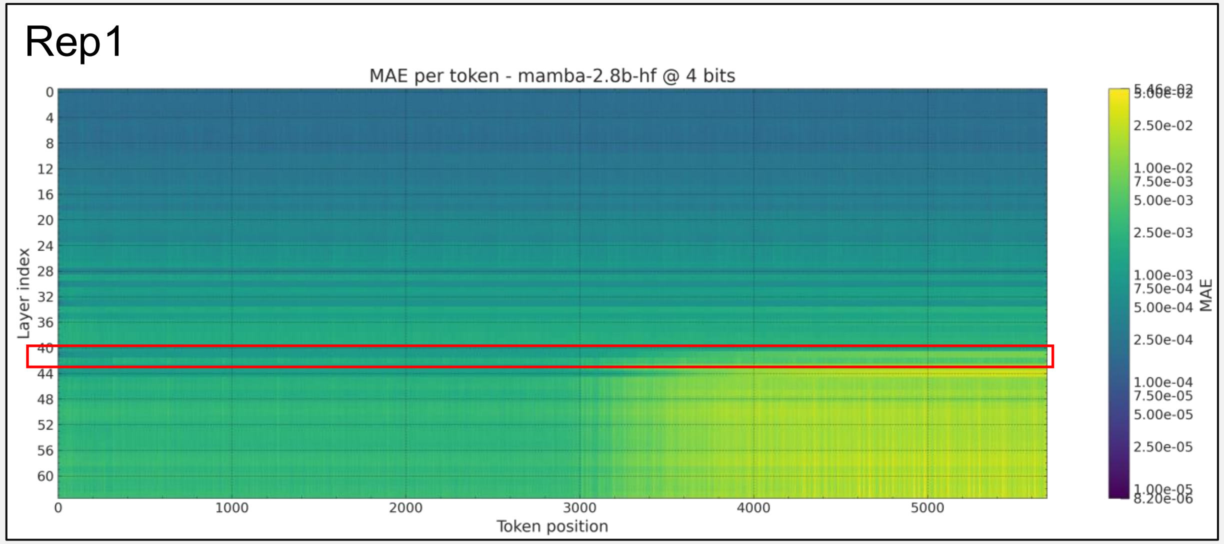
Skipping quantization of one layer

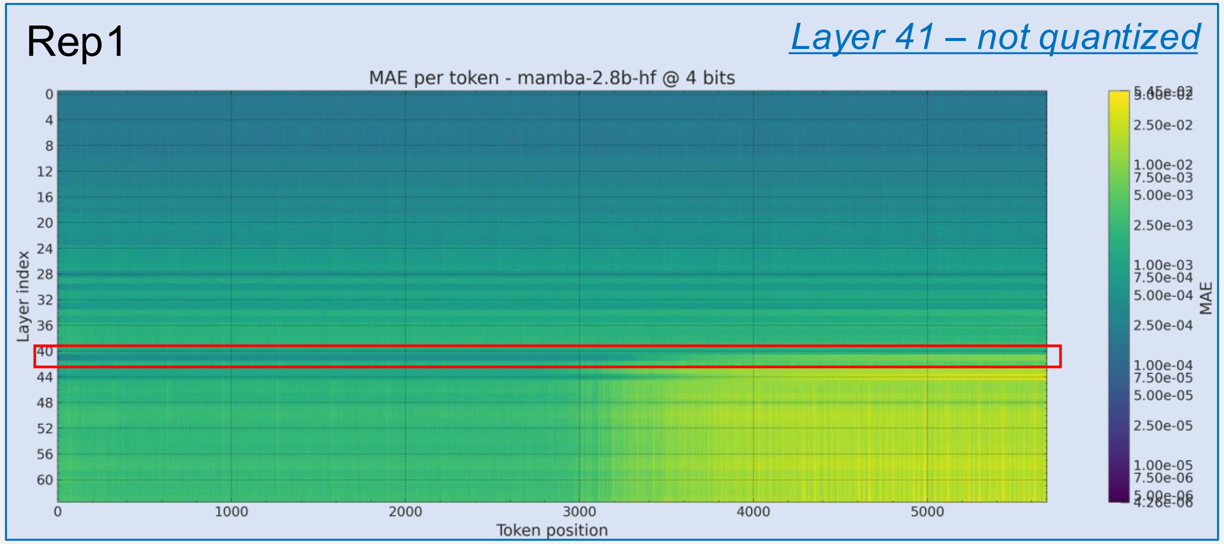




- Considering the maximum absolute difference instead of the relative, there is almost no difference in the patterns.

Skipping quantization of one layer





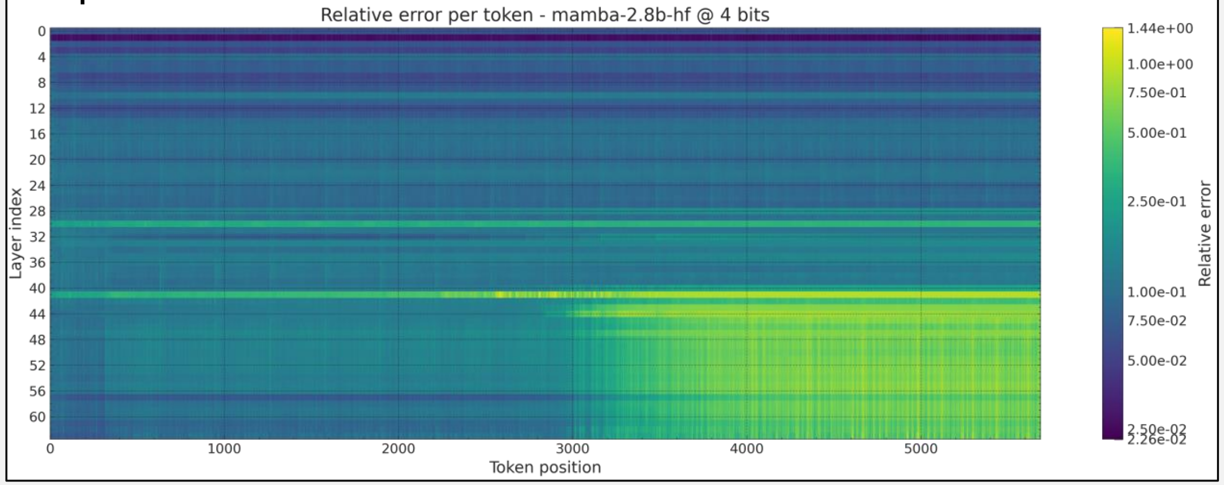
- Considering the mean absolute difference instead of the relative, there is no difference in the patterns.

The difference at the last layers between the quantized and the floating point models is not due to the quantization error introduced by layer 41

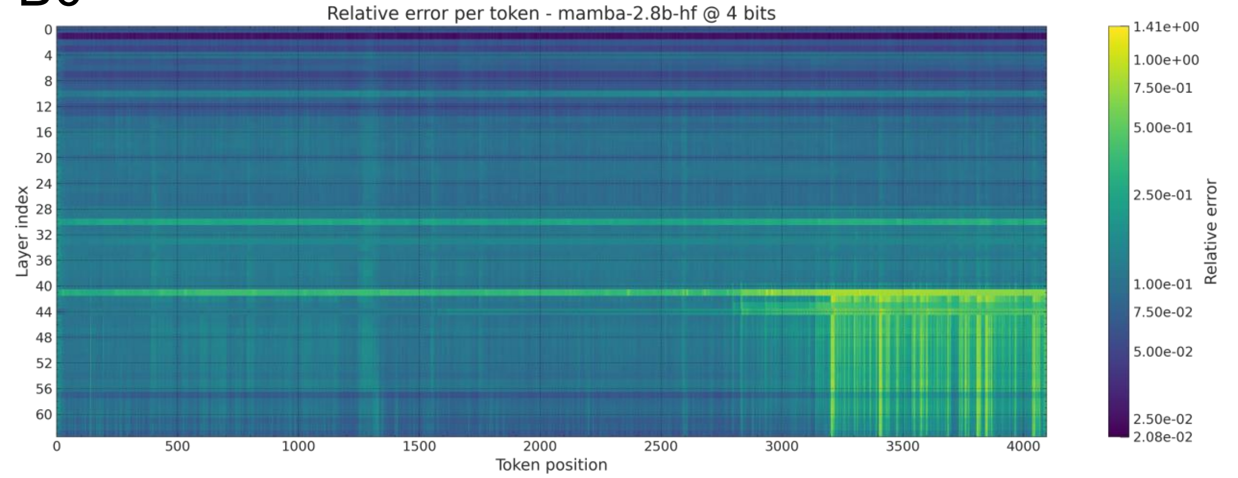
Skipping quantization of early layers

- From past observations:

Rep1



B0

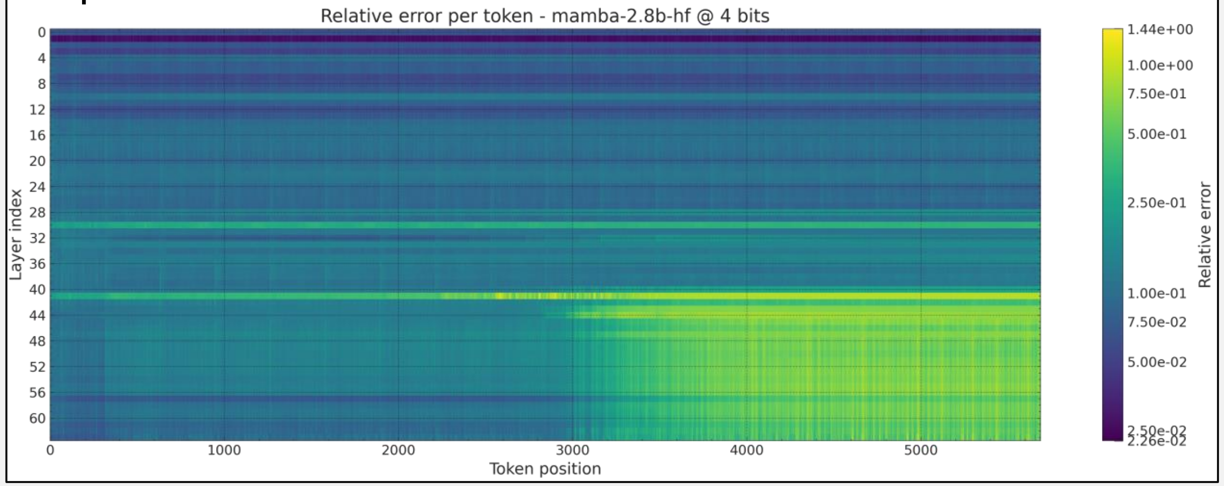


➢ What is the effect of skipping quantization of all the layers up to block 41?

State drift

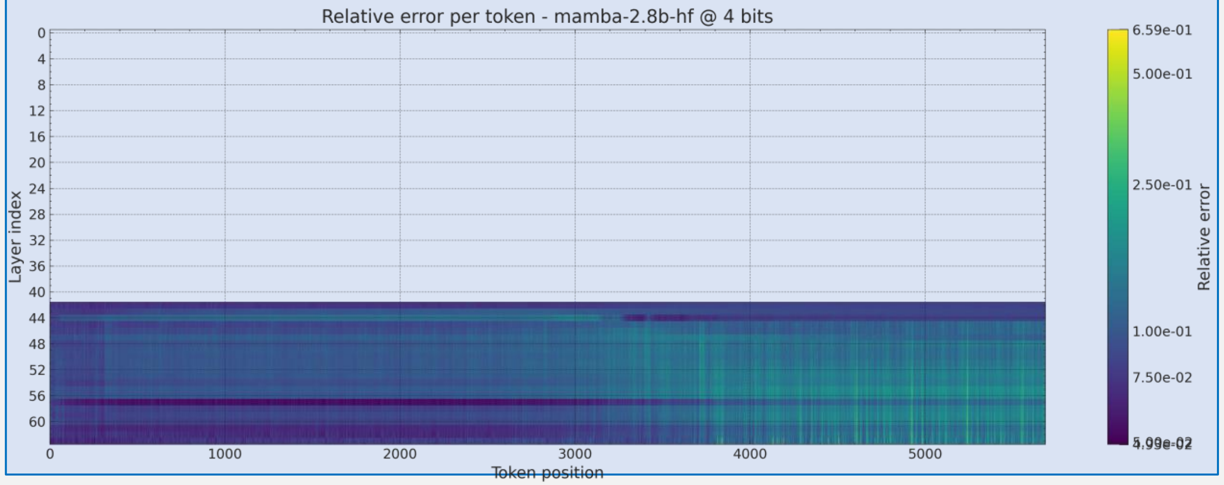
Skipping quantization of early layers

Rep1

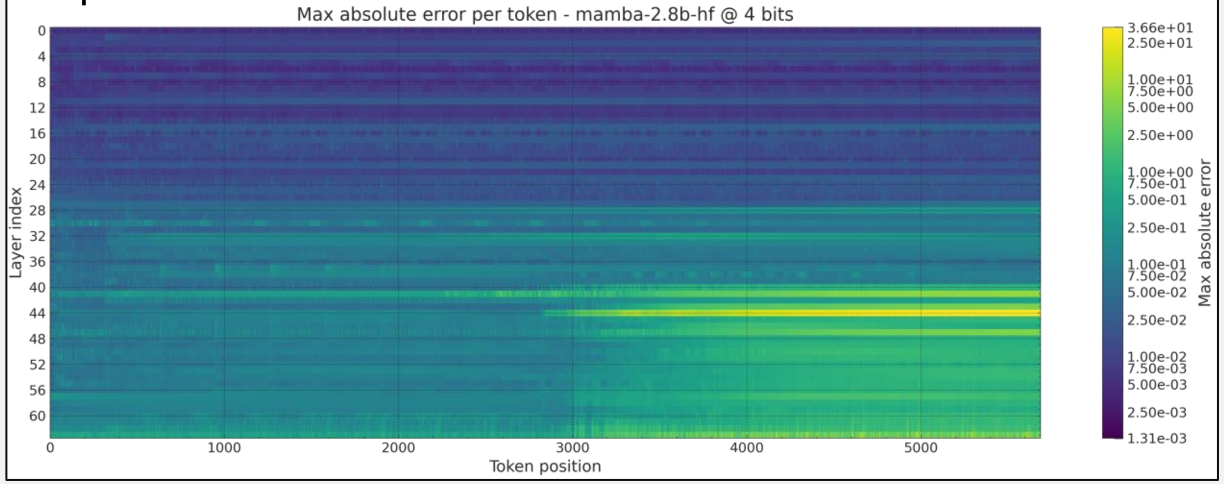


Rep1

Layers up to 41 – not quantized

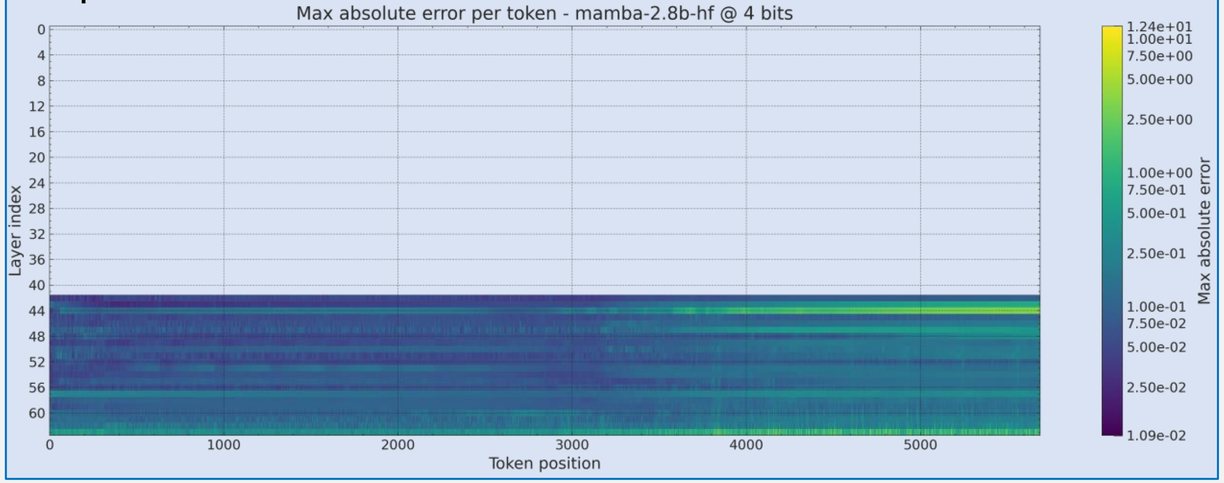


Rep1

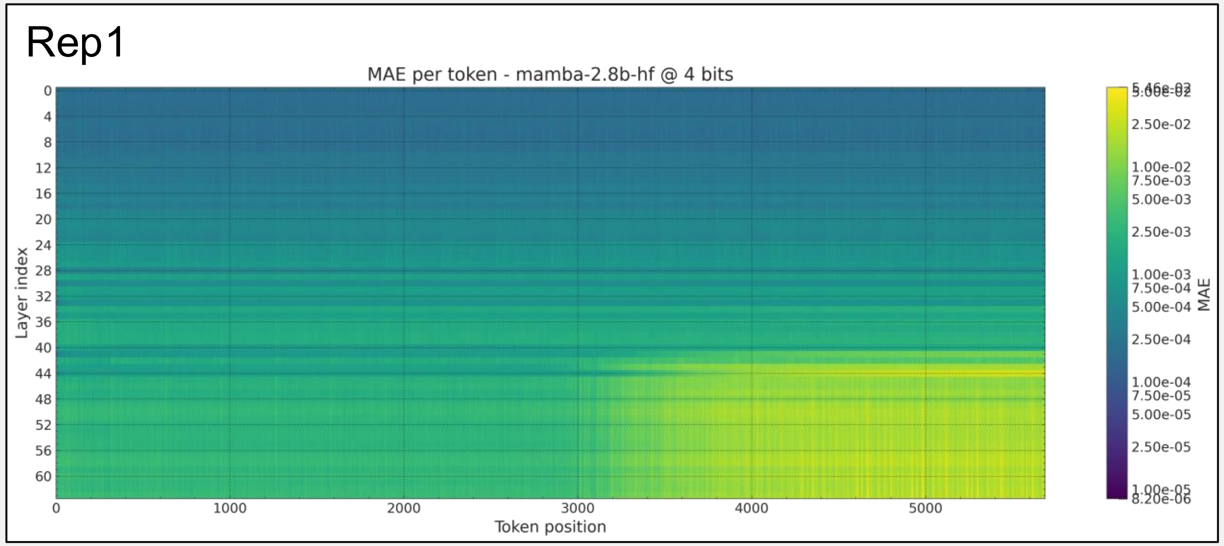


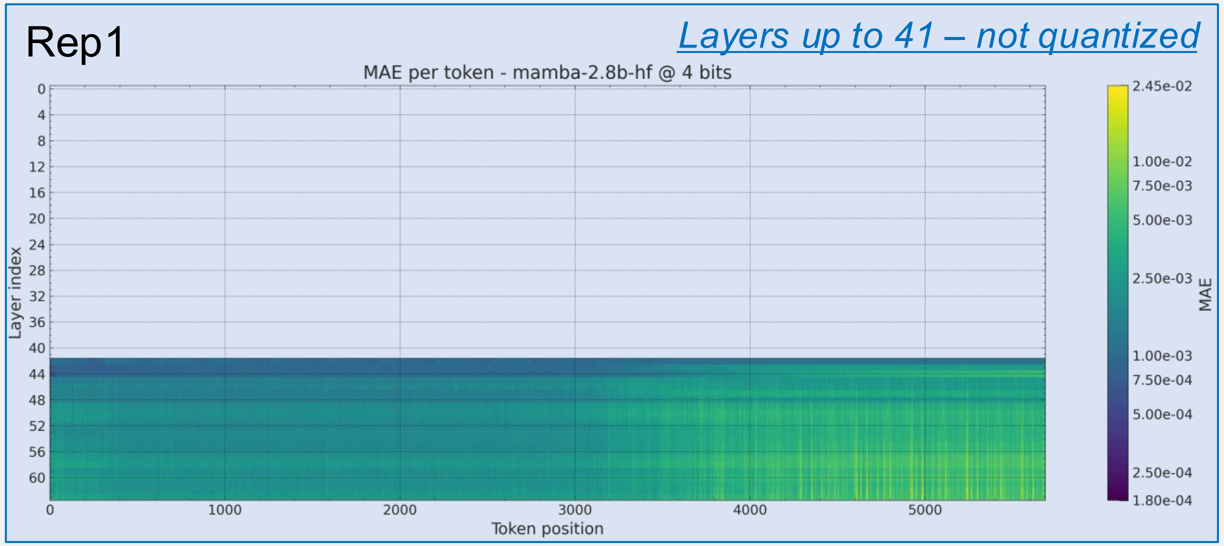
Rep1

Layers up to 41 – not quantized



Skipping quantization of early layers



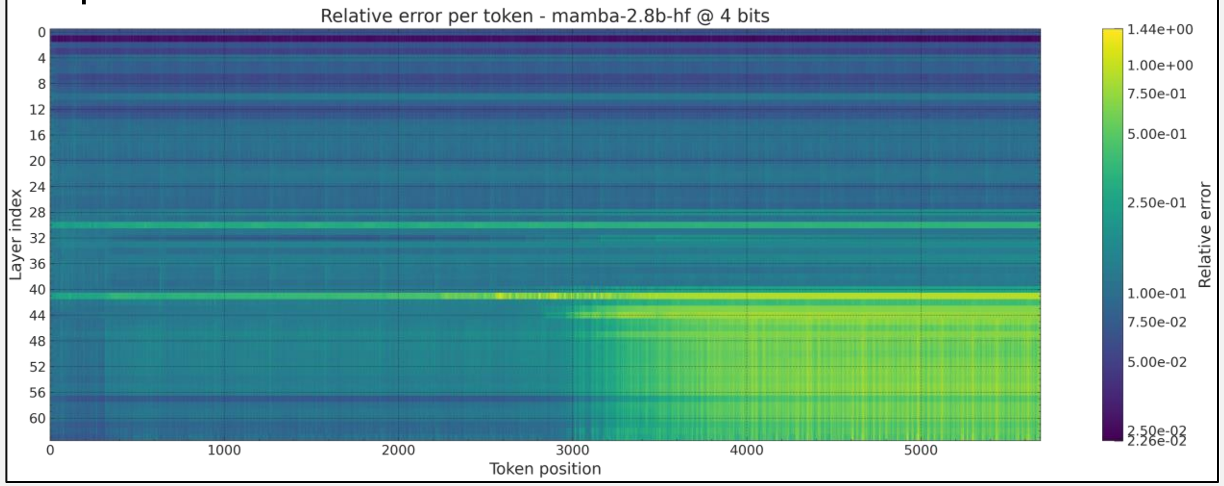


- 0.5x MAE at the later timesteps

State drift

Skipping quantization of last layers

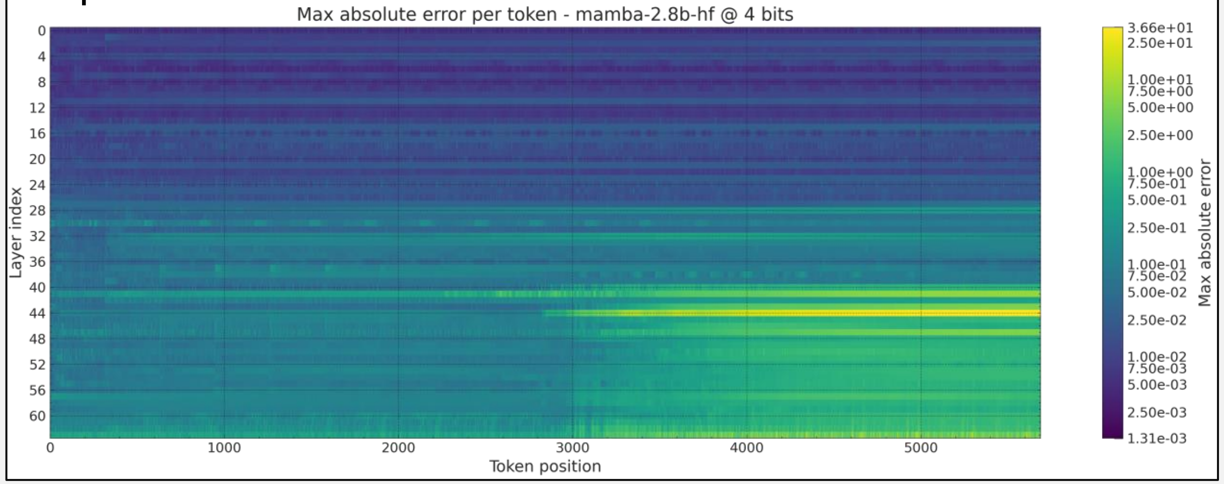
Rep1



Rep1

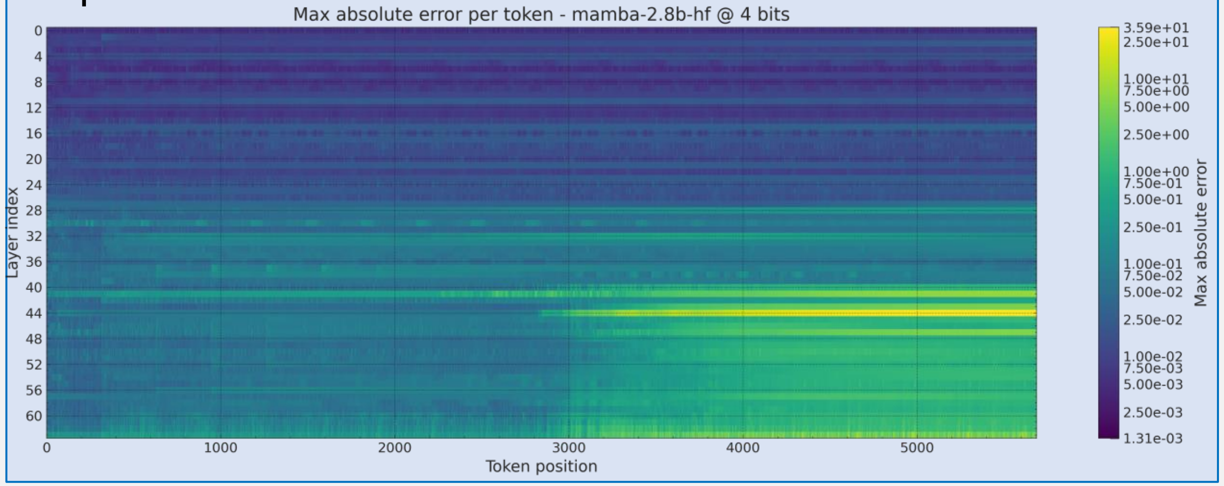
Layers from 42 not quantized

Rep1

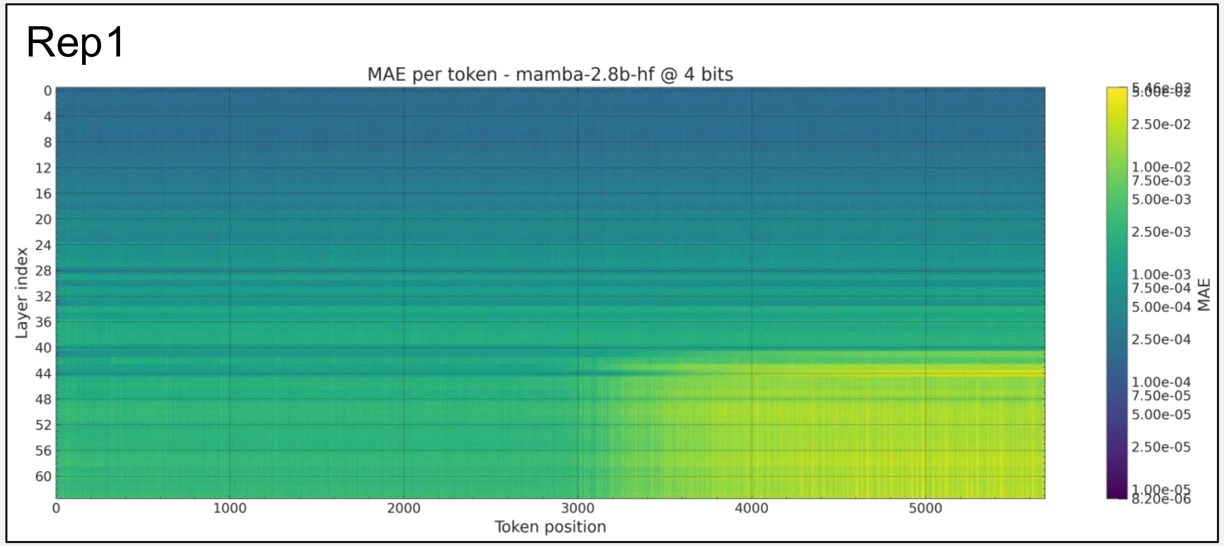


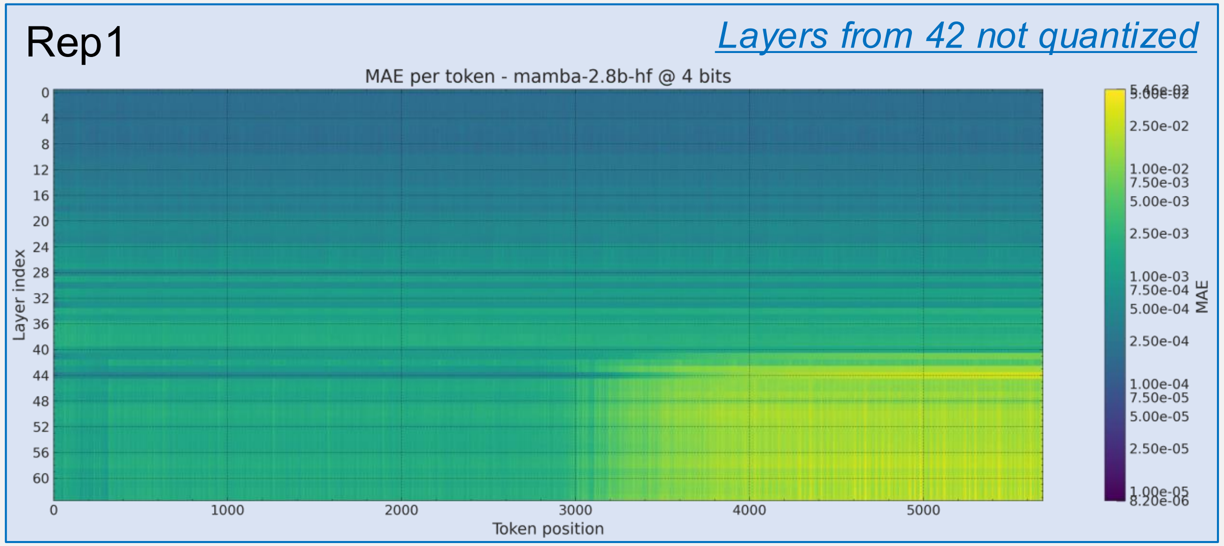
Rep1

Layers from 42 not quantized



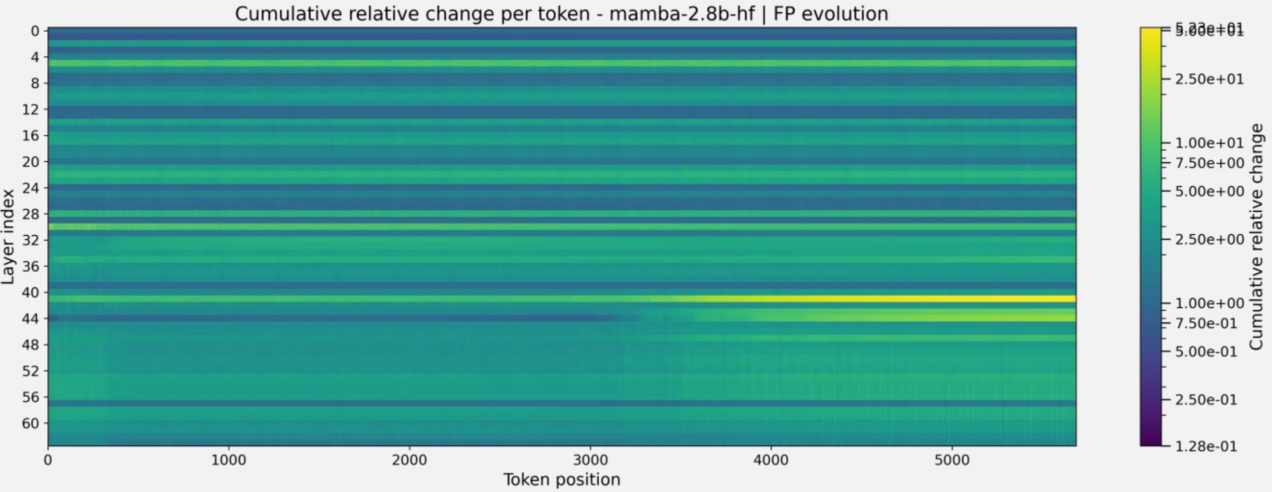
Skipping quantization of last layers

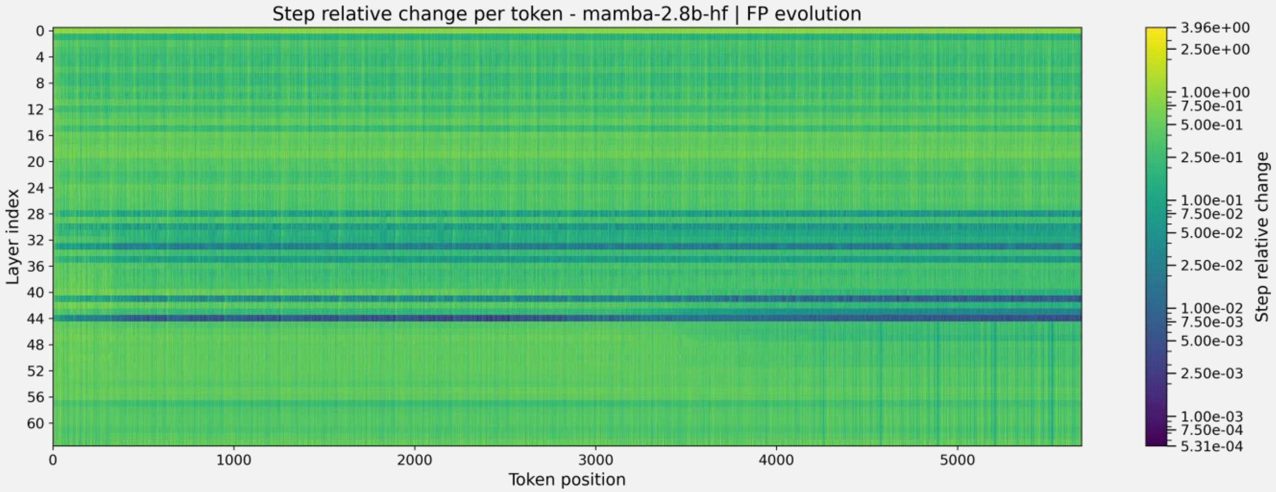




- Negligible change: the state still drifts from the floating point, even if those layers are not quantized → **the change effect comes from the layers before**

FP model state evolution



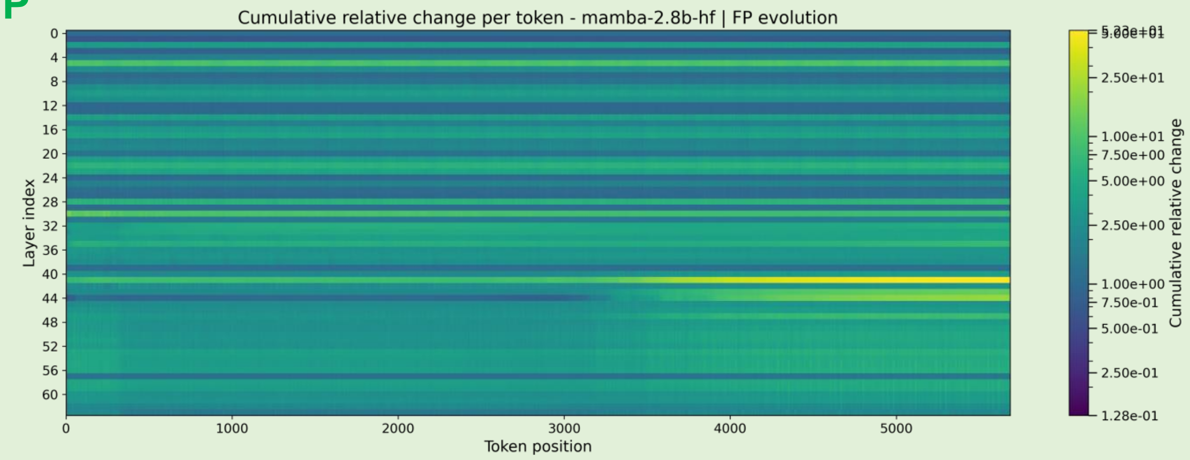


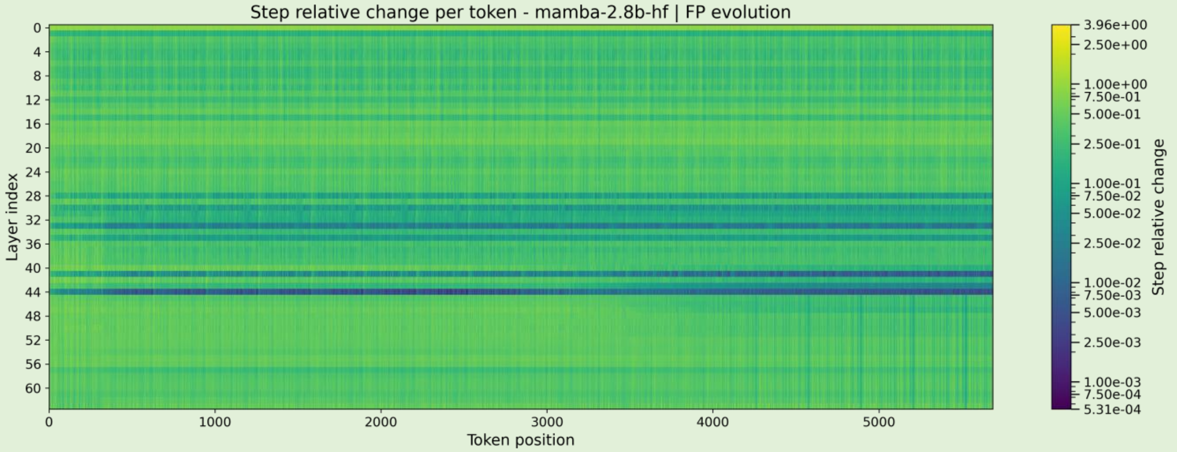
- Channel 41, 44 (and others) change slightly at each step.
    - They end up being extremely different from the start (after token 0)

State drift

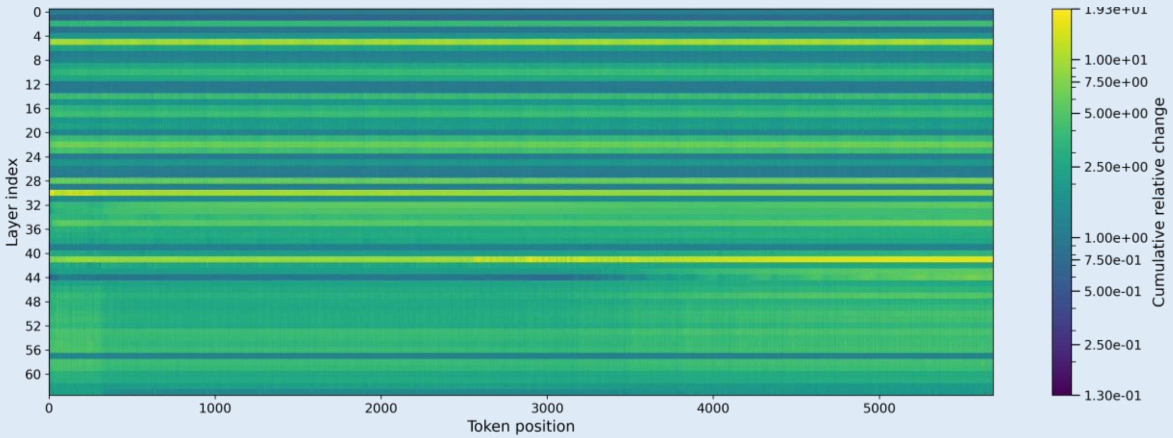
FP vs Q model states evolution

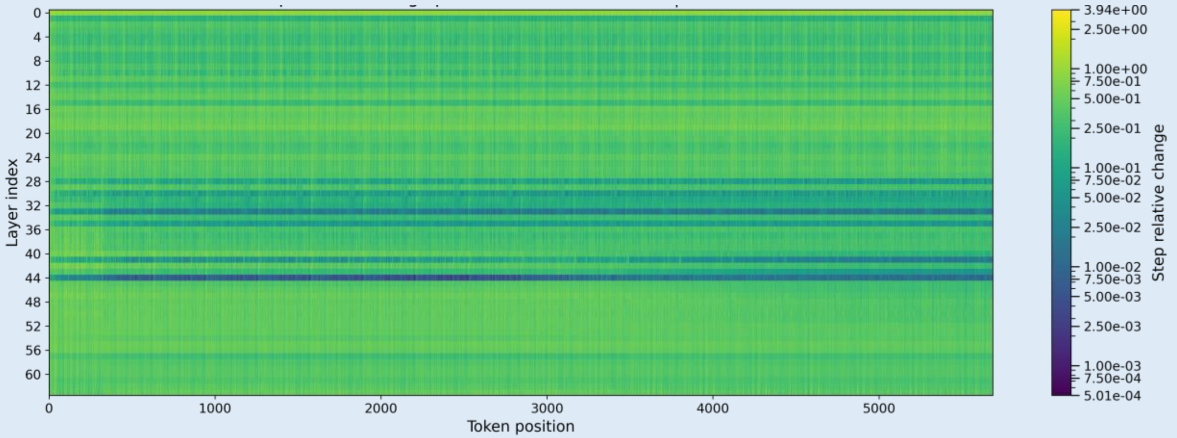
FP

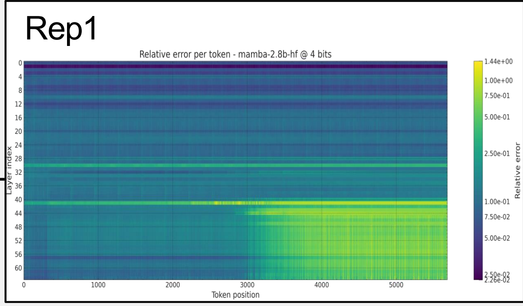




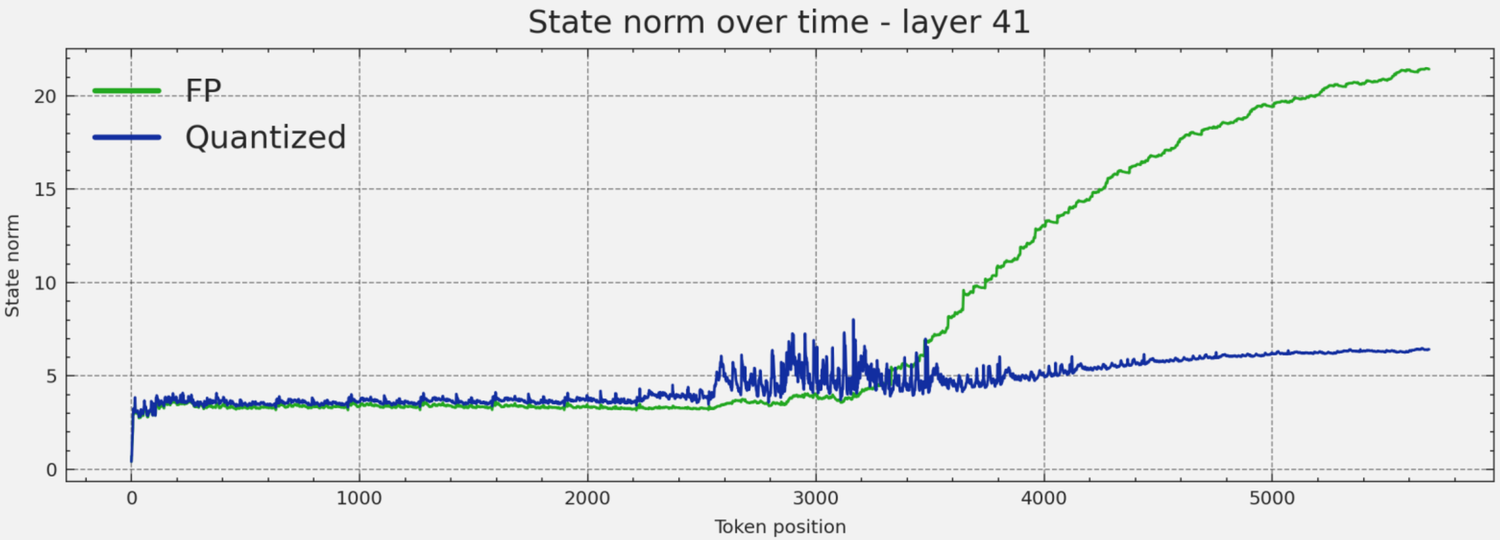
Q





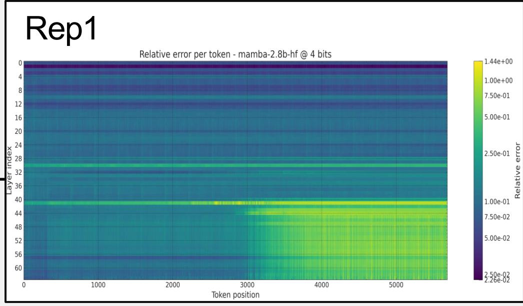


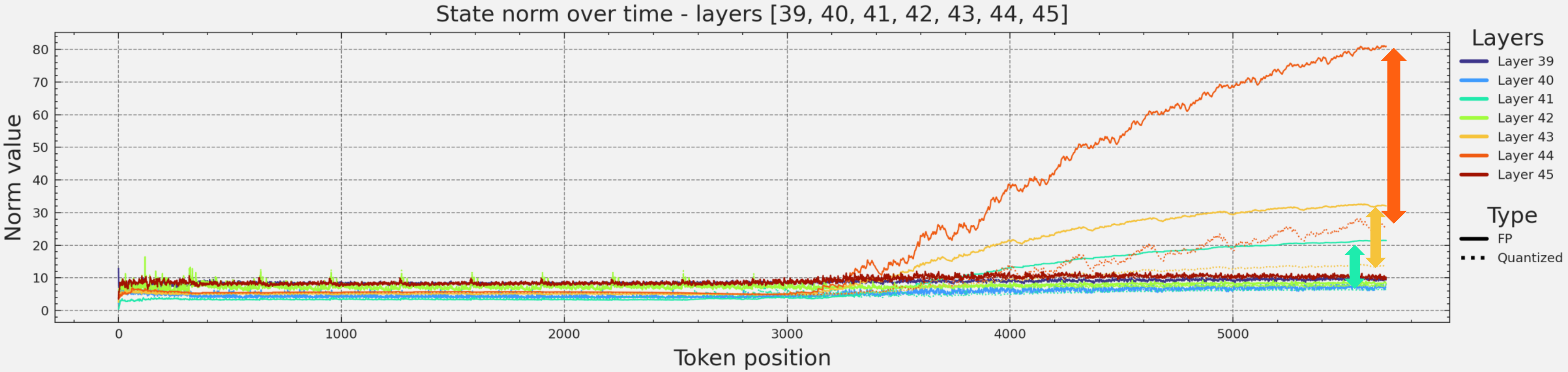
Norm of layers



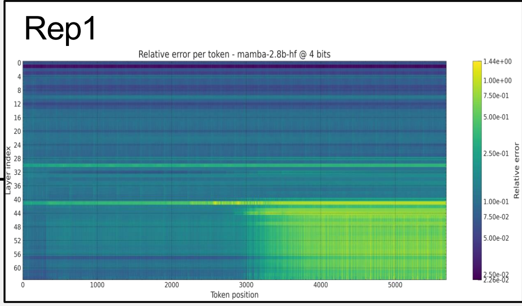
- The floating point model's state norm increases considerably over time, and the quantized model does not follow.

Norm of layers vs cosine sim.





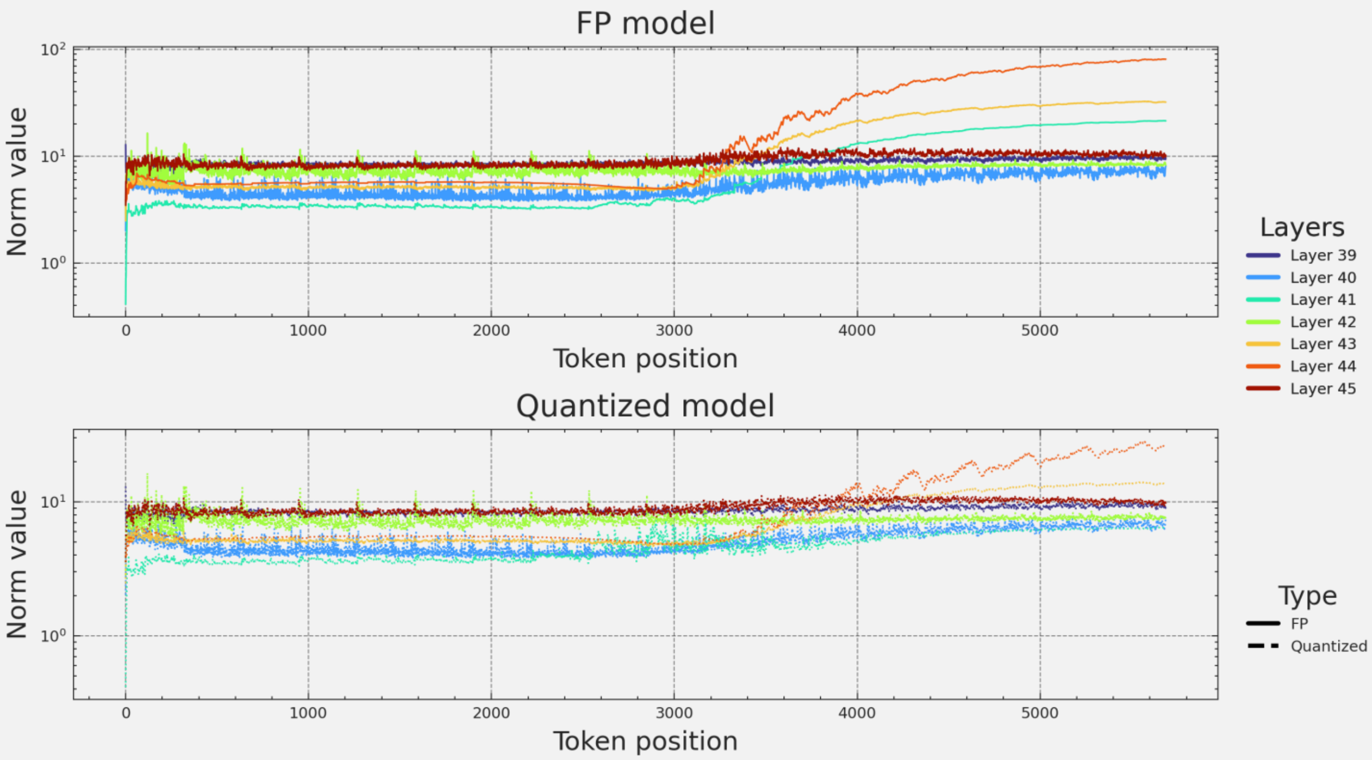
- There is a large gap in state norm magnitude for layers 44, 43, and 41.

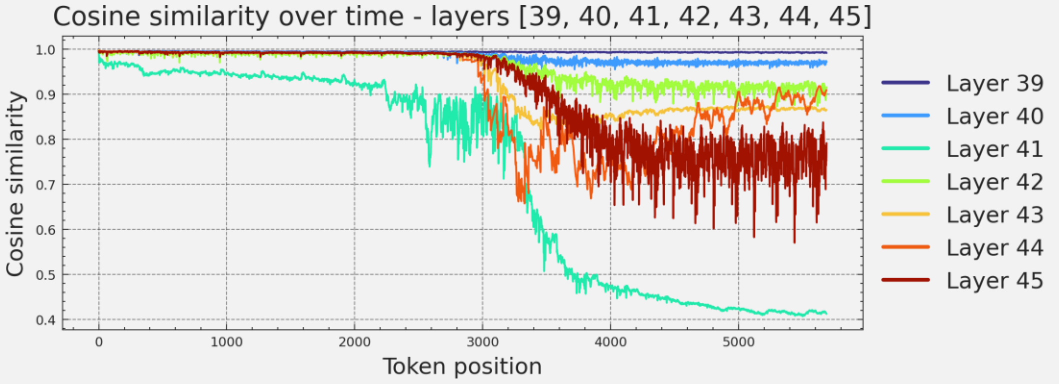


State drift

Norm of layers vs cosine sim.

Separate view of quantized and FP:

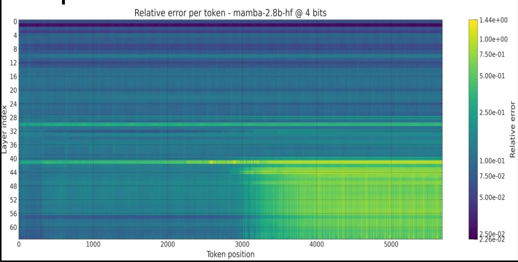




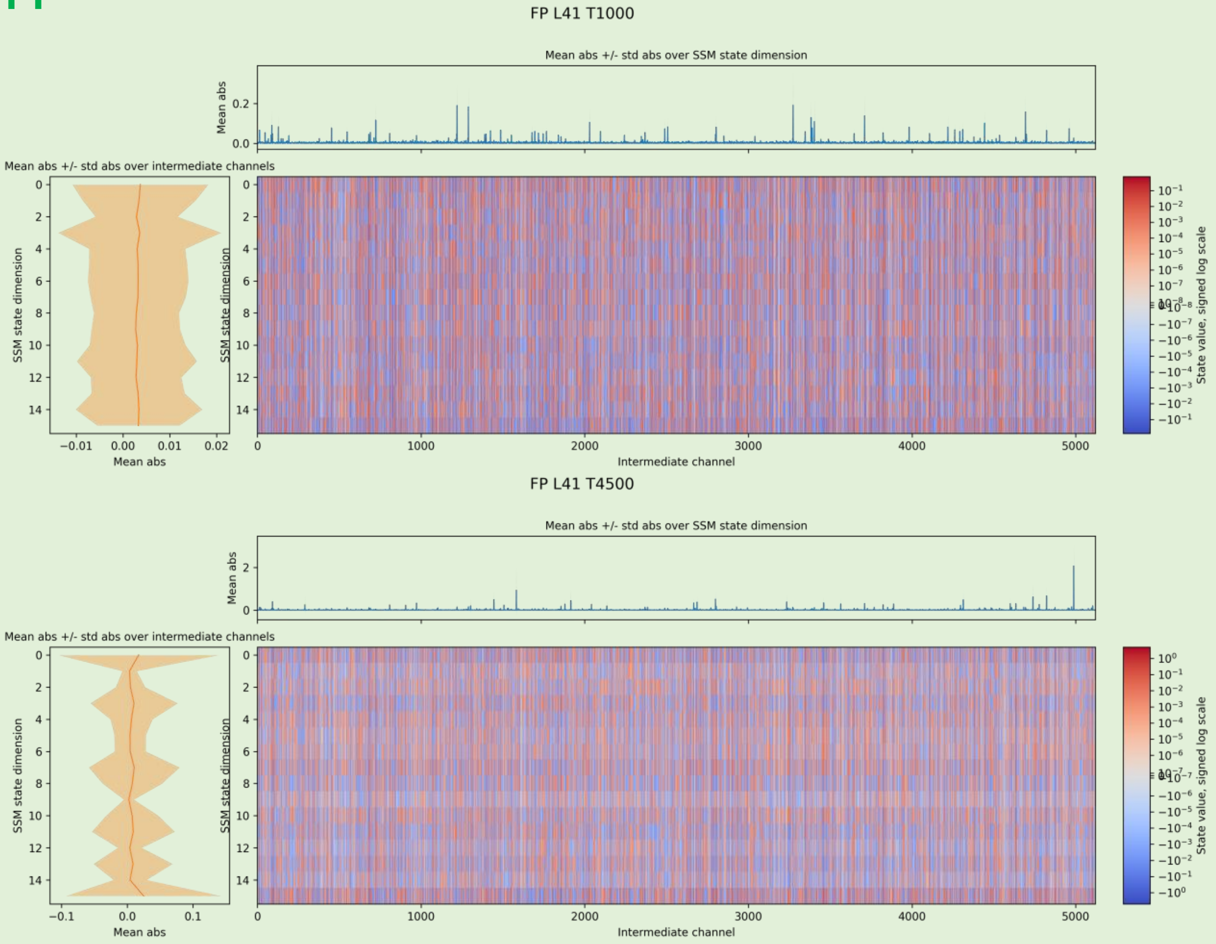
- Layer 41’s has the lowest cosine similarity (FP vs Q), low values

Layer 41 states

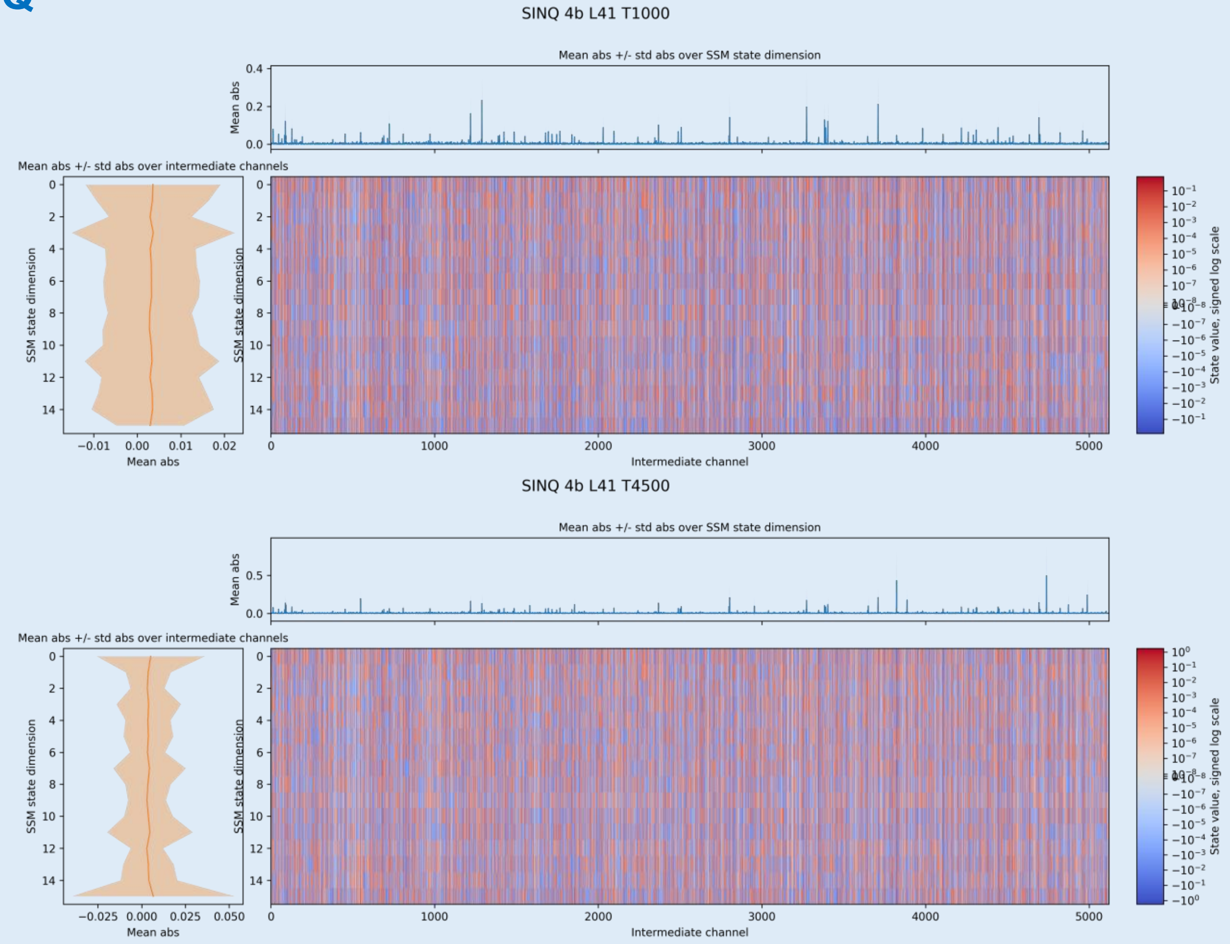
Rep1



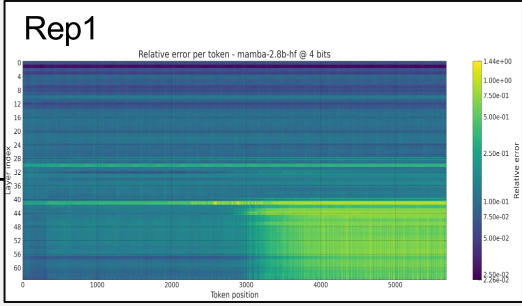
FP

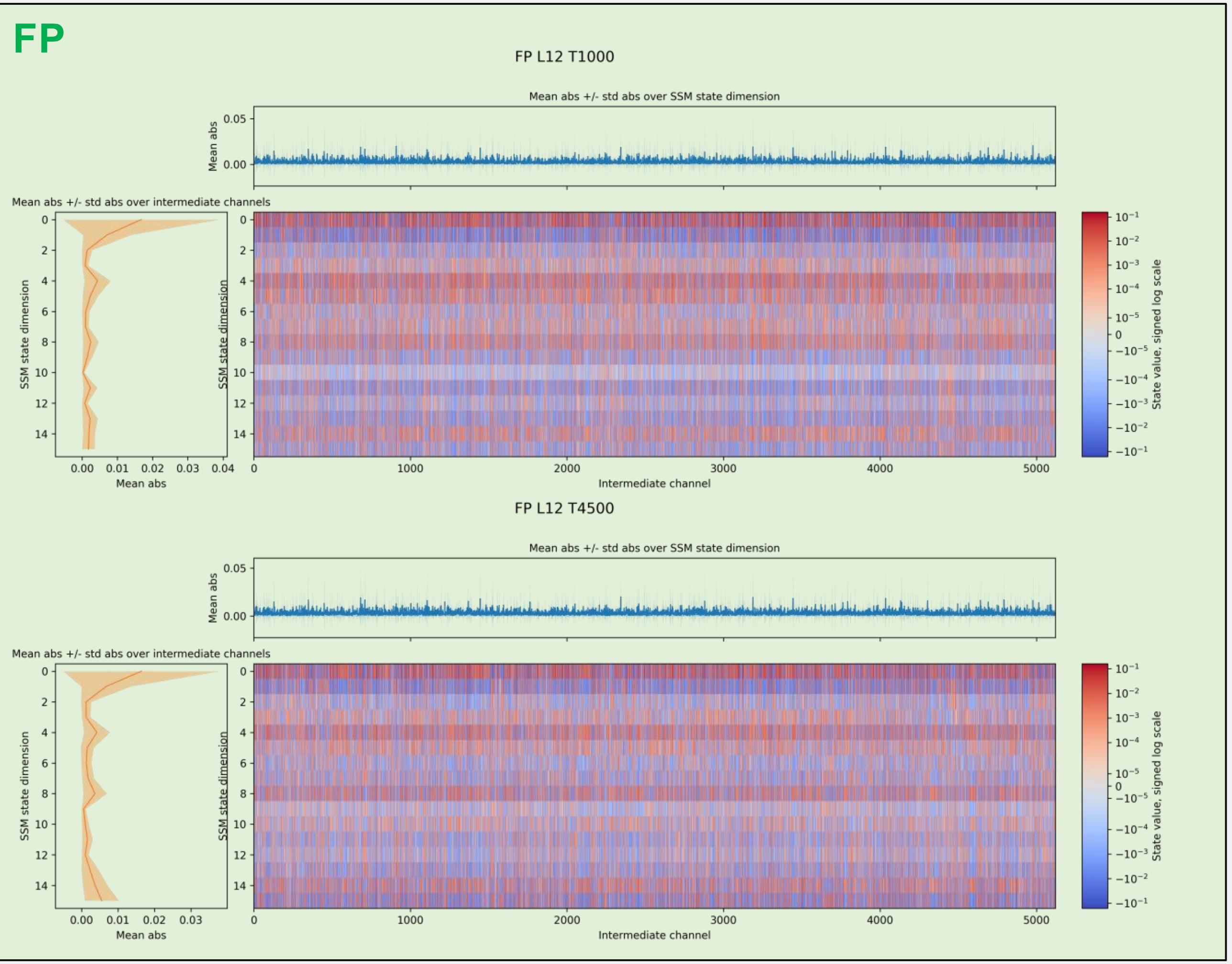


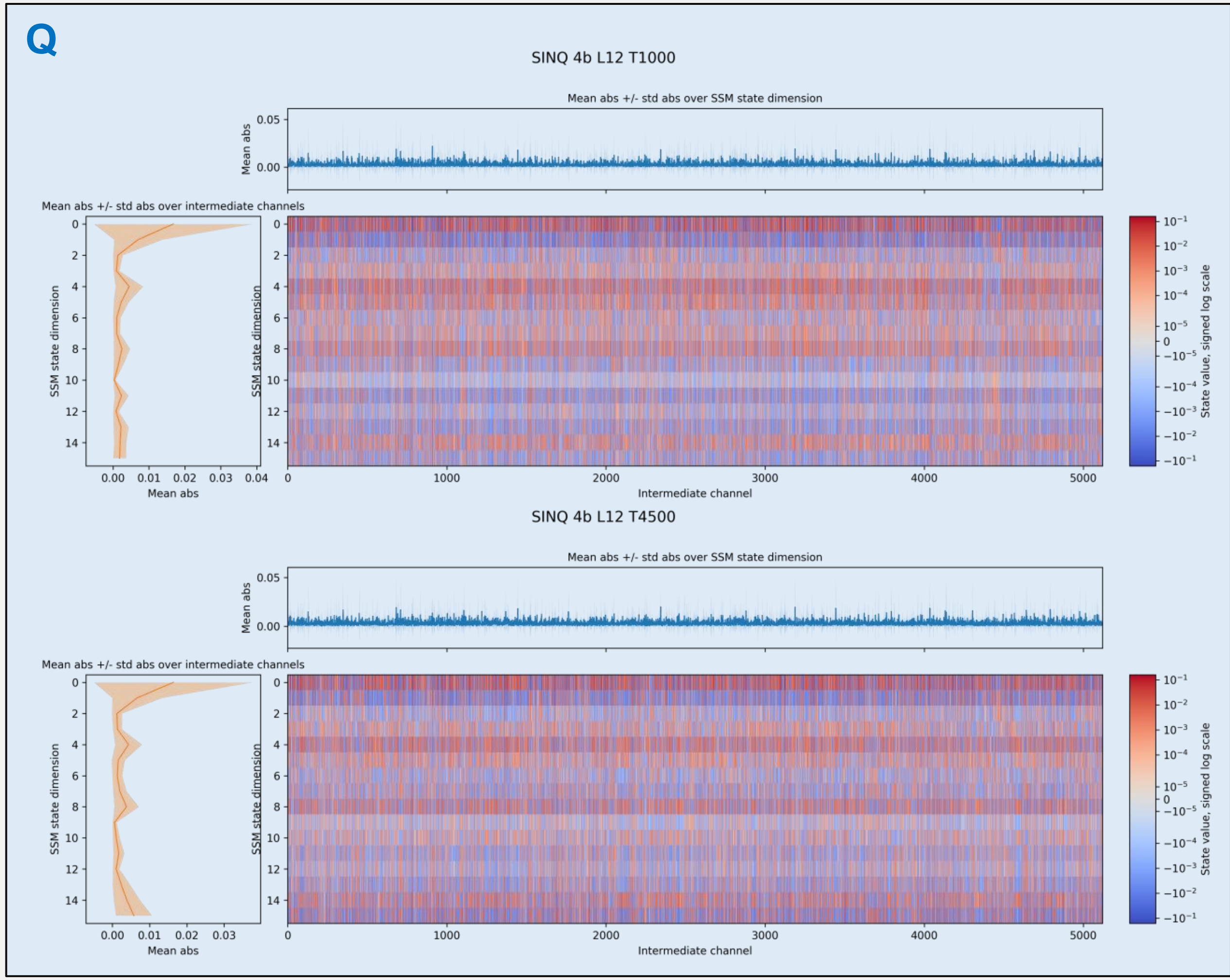
Q



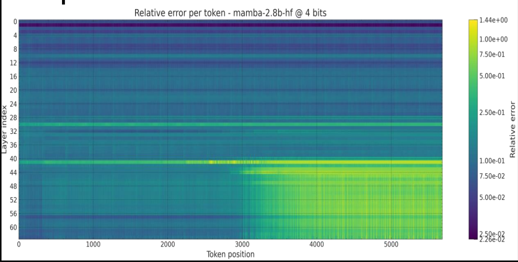
Layer 12 states







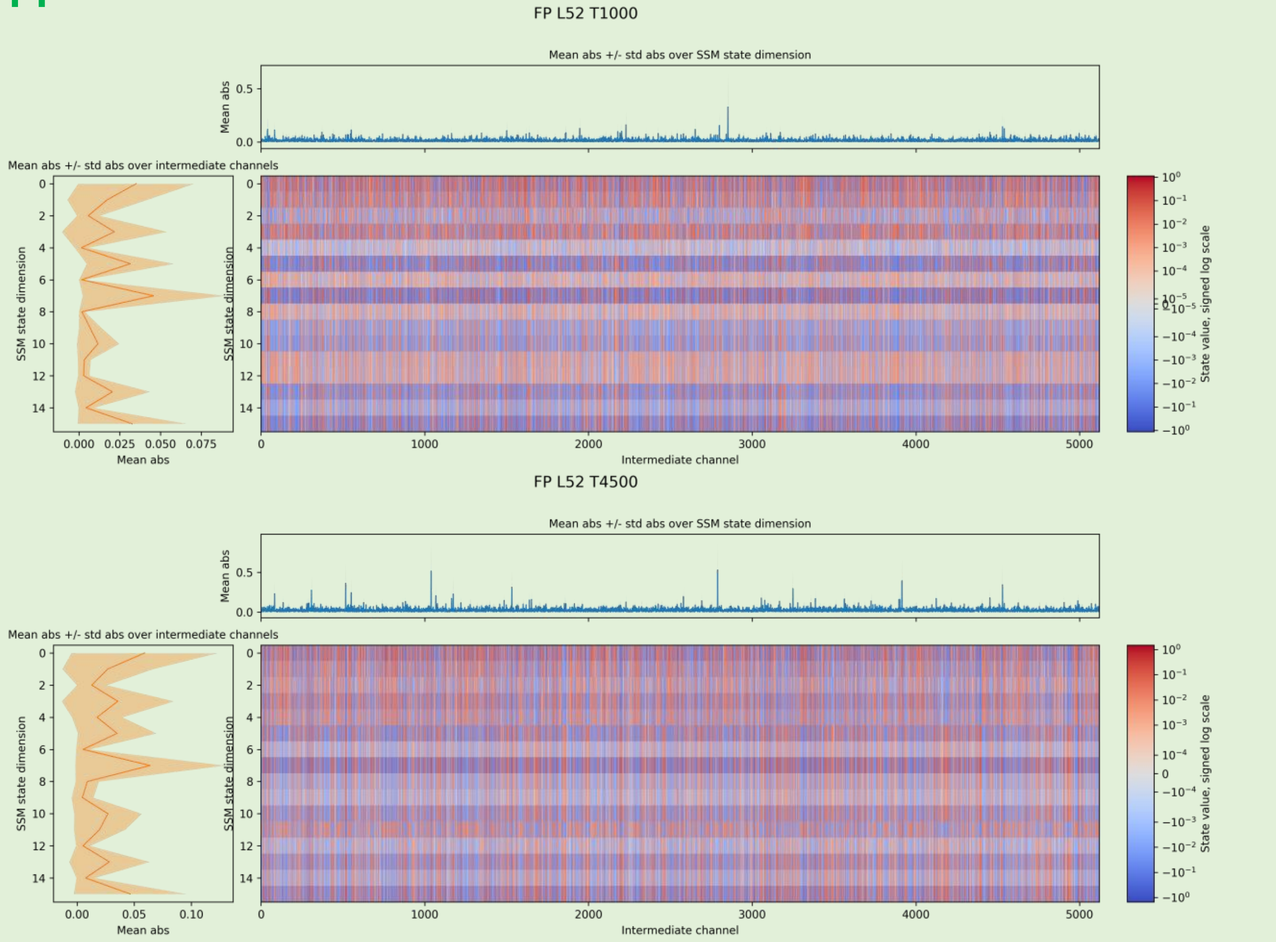
Rep1



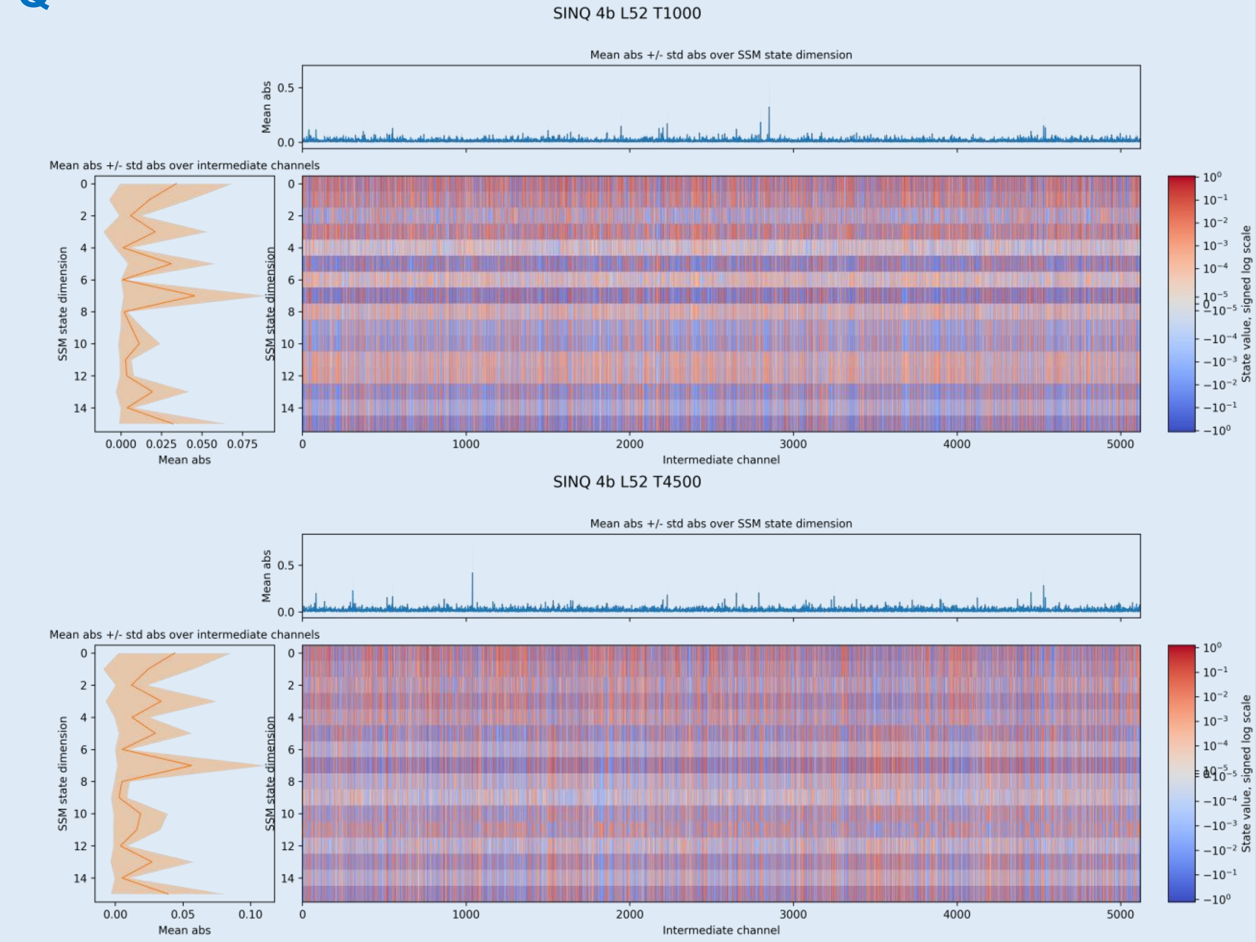
State drift

Layer 52 states

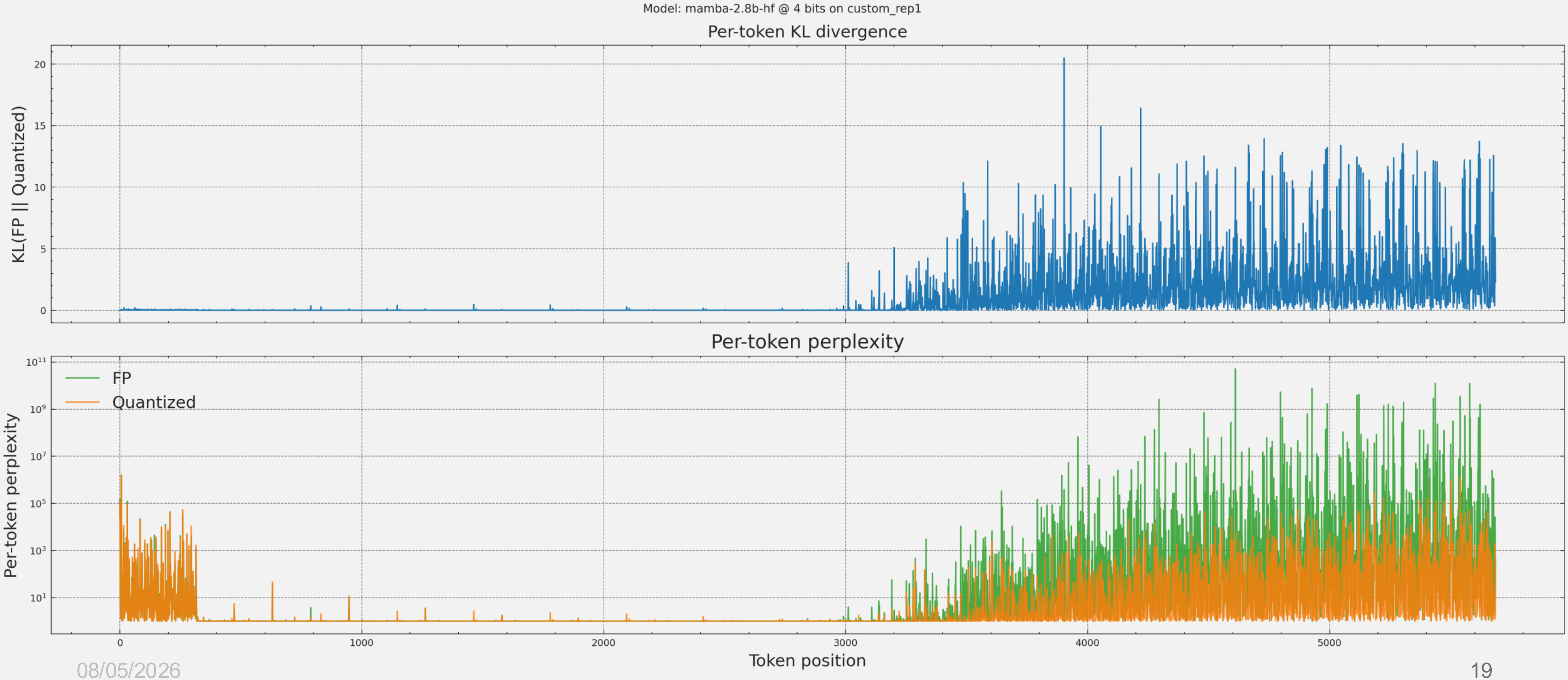
FP



Q



KL divergence & perplexity



State drift

Output with repeated pattern

Original text (from bbc-news dataset): mobiles double up as bus tickets mobiles could soon double up as travel cards with nokia planning to try out a wireless ticket system on german buses. early next year travellers in the city of hanau near frankfurt will be able to pay for tickets by passing their phone over a smart-card reader already installed on the buses. passengers will need to own a nokia 3220 handset which will have a special shell attached to it. the system would reduce queues and make travelling easier said nokia. transport systems around the world are seeing the advantage of using ticketless smartcards. using a mobile phone is the next step said gerhard romen head of market development at nokia. the ticketless trial will start early in 2005 and people will also be able to access transport information and timetables via their phones. nokia has worked with electronics giant philips to develop a shell for the mobile phone that will be compatible with hanau s existing ticketing system. the system opens up possibilities for mobile devices to be interact with everyday environments said mr romen. it could be used in shops to get product information at bus-stops to get information about the next bus or for example by being passed over an advert of a rock star to find out details of concerts or get ringtones he told the bbc news website. he is confident that the trial being run in germany could be extended to transport systems in other countries. the technology offers access to a lot of services and makes it easy to get the information you want he said. **[x18]**

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✓ print(data["sequences"][0]["fp_predicted_text"]) ···

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✓ print(data["sequences"][0]["quantized_predicted_text"]) ···

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Layer 41

State update in the SSM block: `ssm_state = discrete_A[:, :, i, :] * ssm_state + deltaB_u[:, :, i, :]`

Where: `discrete_A = torch.exp(A[None, :, None, :] * discrete_time_step[:, :, :, None])`

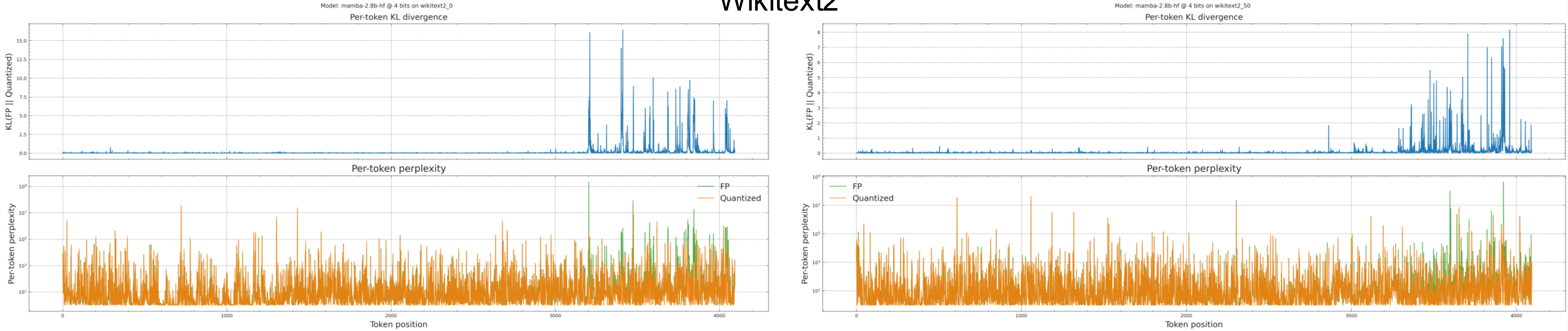
- In block 41, 40% of values of discrete_A are exactly equal to 1. This is due to the 98% values of `discrete_time_step` being less than 1e-3 (median ~7e-8).

- The tokens which have the lowest discrete_A values (i.e, allowing updates of the state) are at positions [0,1,2,3,7,8,9,3330,3477,3646,3793,3962]

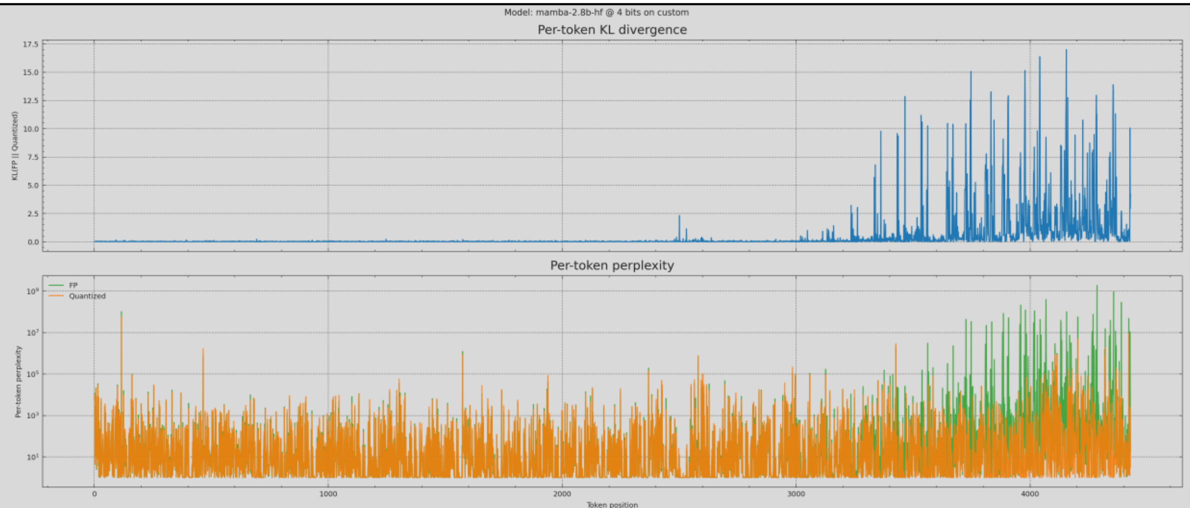
Original text (from bbc-news dataset): mobiles double up as bus tickets mobiles could soon double up as travel cards with nokia planning to try out a wireless ticket system on german buses. early next year travellers in the city of hanau near frankfurt will be able to pay for tickets by passing their phone over a smart-card reader already installed on the buses. passengers will need to own a nokia 3220 handset which will have a special shell attached to it. the system would reduce queues and make travelling easier said nokia. transport systems around the world are seeing the advantage of using ticketless smartcards. using a mobile phone is the next step said gerhard romen head of market development at nokia. the ticketless trial will start early in 2005 and people will also be able to access transport information and timetables via their phones. nokia has worked with electronics giant philips to develop a shell for the mobile phone that will be compatible with hanau s existing ticketing system. the system opens up possibilities for mobile devices to be interact with everyday environments said mr romen. it could be used in shops to get product information at bus-stops to get information about the next bus or for example by being passed over an advert of a rock star to find out details of concerts or get ringtones he told the bbc news website. he is confident that the trial being run in germany could be extended to transport systems in other countries. the technology offers access to a lot of services and makes it easy to get the information you want he said. **[x18]**

KL div. & perplexity: other prompts

Wikitext2



Custom, not repeated



Q-Mamba [1]

[1] Chen et al., "*Q-Mamba: Towards more efficient Mamba models via post-training quantization*", Findings of ACL 2025

- Quantization on both **linear projections** of Mamba2 models and **state caches** (idea is to load and dequantize them before computation at the next timestep).
- They observe that states exhibit outliers in both the state and the channel dimension, and they propose to use **separate scales for the state and the channel dimensions**.
  - The channel scales are computed as the square root of the mean of the channel values.
  - The state scales are computed as the maximum values of the ratio between absolute state values and the channel scale.
- They **fine-tune the linear layers** (except input and output projection layers) on a small dataset of 128 samples of length of 2048, in floating point. They also apply SmoothQuant with per-token quantization.
- They achieve 50% memory reduction with only ~2% accuracy loss on zero-shot tasks for Mamba2-2.7B using 8-bit weights/activations and 4-bit state caches.

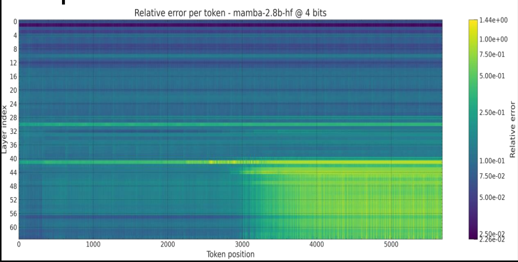
Next steps

- Analysis of the quantization of linear layers of block 41 of Mamba-2.8b to verify whether we can identify the source of noise.
- Why something changes after step 3000?

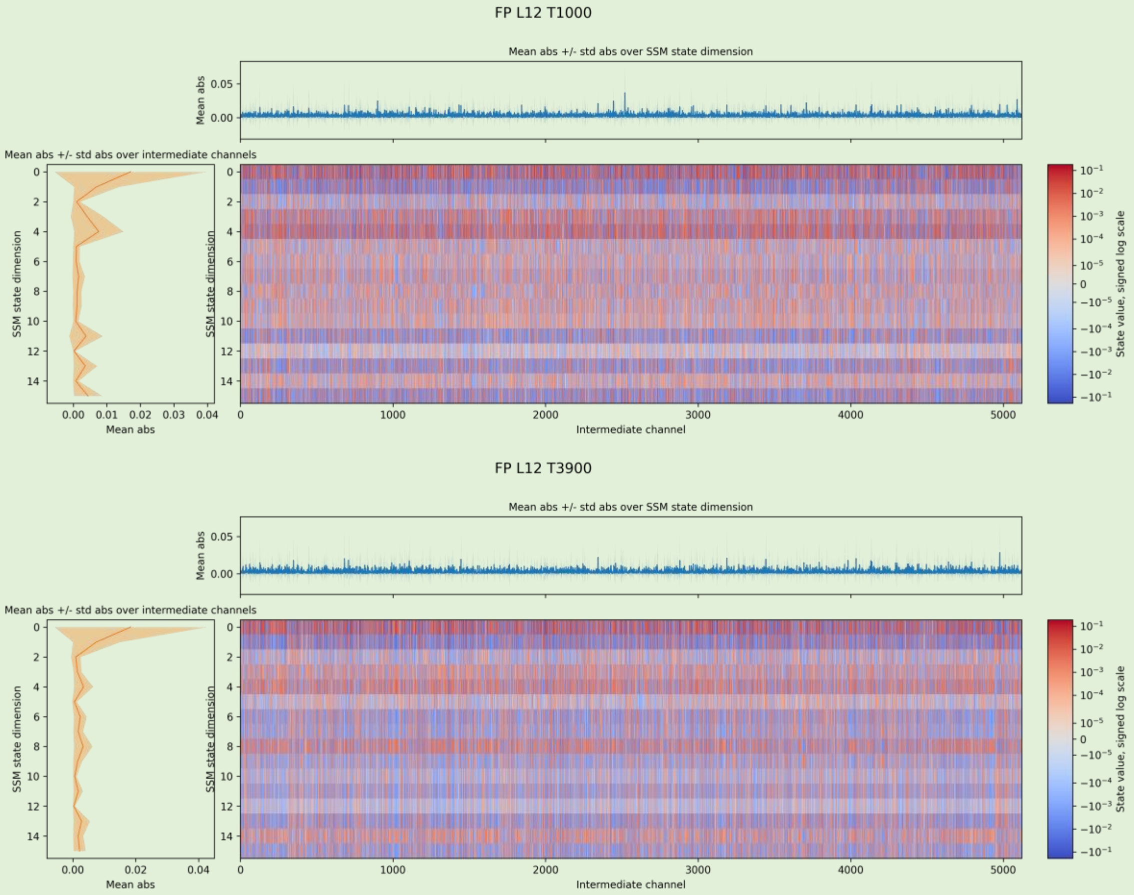
State drift

Layer 12 states – Wikitext2_0

Rep1



FP



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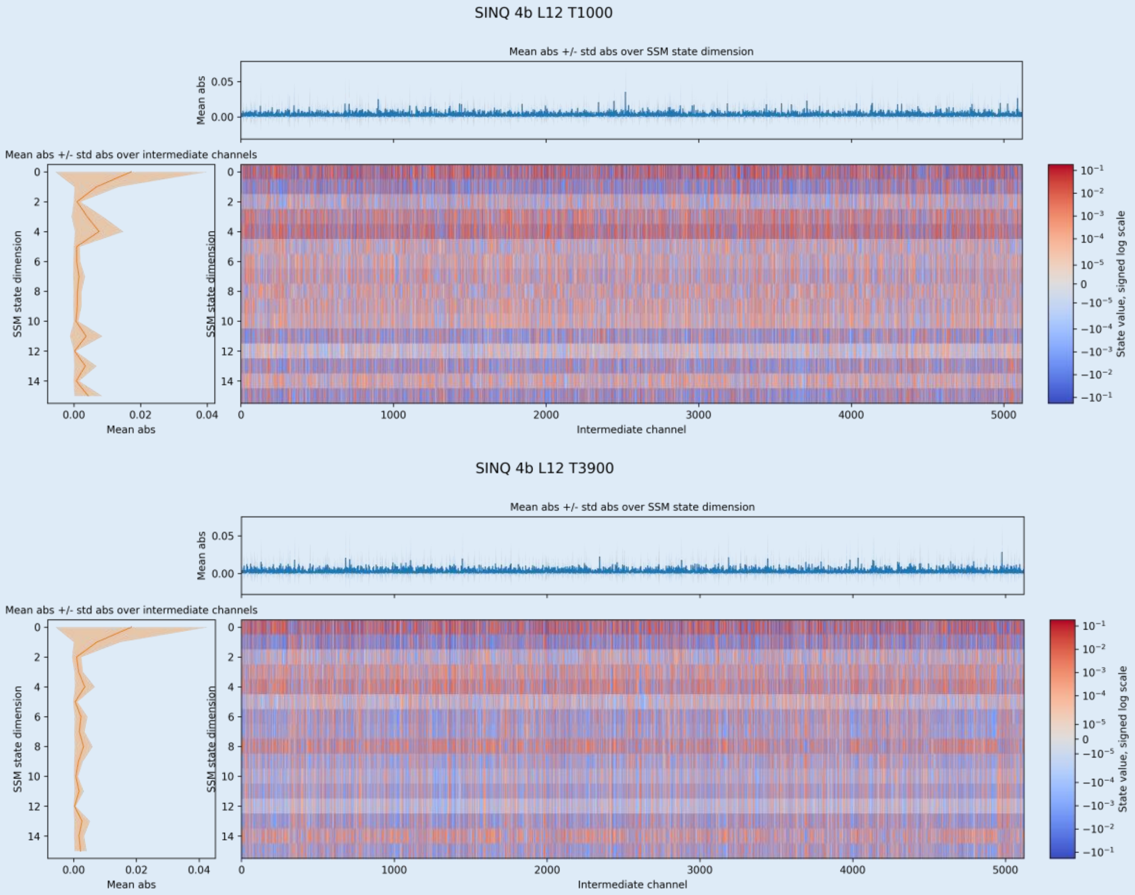
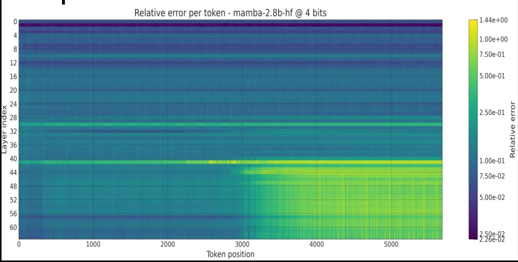


State drift

Layer 41 states – Wikitext2_0

Rep1



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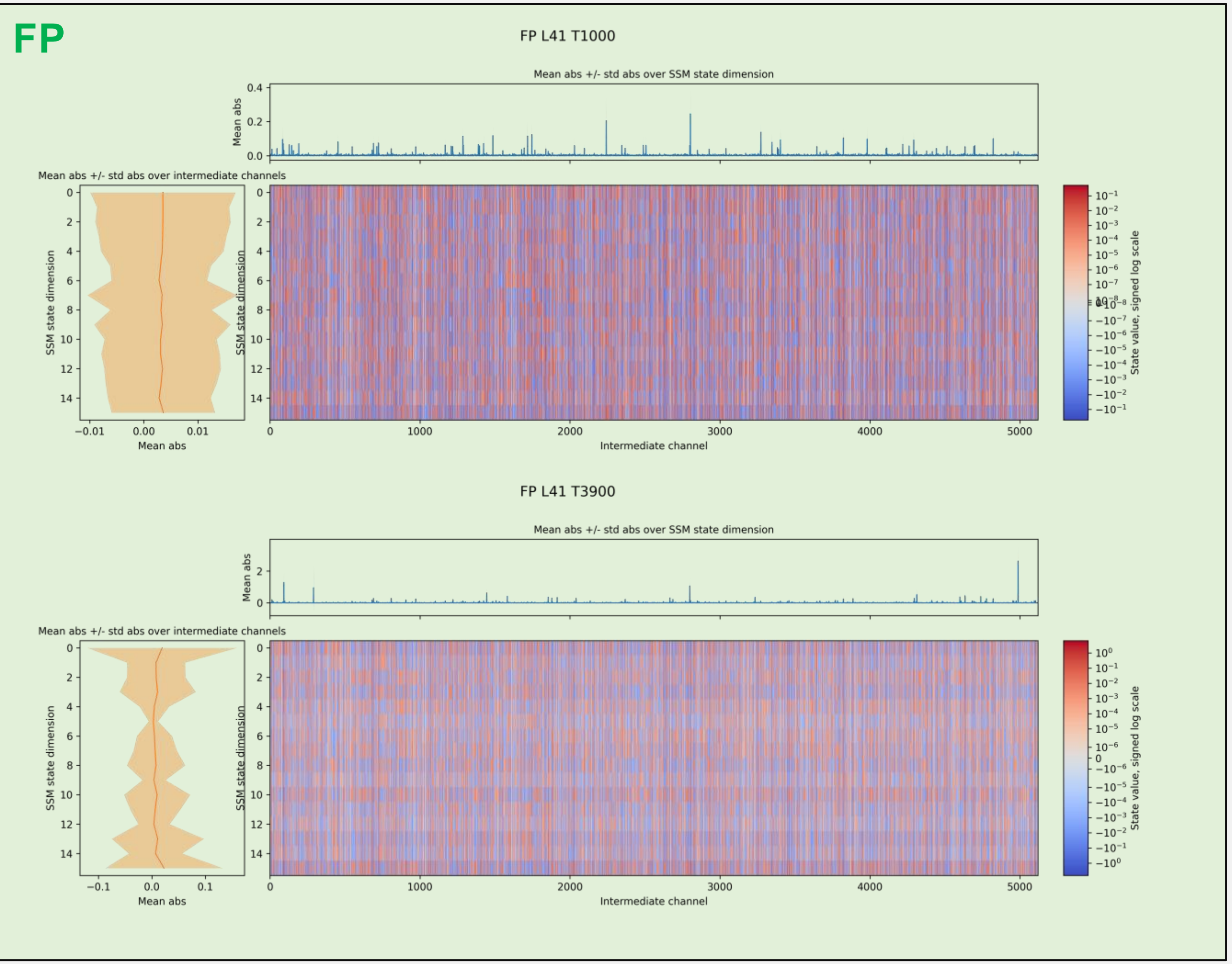

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